

Identification of pharmacological inducers of a reversible hypometabolic state for whole organ preservation

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Abstract

Drugs that induce reversible slowing of metabolic and physiological processes would have great value for organ preservation, especially for organs with high susceptibility to hypoxia-reperfusion injury, such as the heart. Using whole-organism screening of metabolism, mobility, and development in *Xenopus*, we identified an existing drug, SNC80, that rapidly and reversibly slows biochemical and metabolic activities while preserving cell and tissue viability. Although SNC80 was developed as a delta opioid receptor activator, we discovered that its ability to slow metabolism is independent of its opioid modulating activity as a novel SNC80 analog (WB3) with almost 1,000 times less delta opioid receptor binding activity is equally active. Metabolic suppression was also achieved using SNC80 in microfluidic human organs-on-chips, as well as in explanted whole porcine hearts and limbs, demonstrating the cross-species relevance of this approach and potential clinical relevance for surgical transplantation. Pharmacological induction of physiological slowing in combination with organ perfusion transport systems may offer a new therapeutic approach for tissue and organ preservation for transplantation, trauma management, and enhancing patient survival in remote and low-resource locations.

eLife assessment

This **important** study provides a new way to enhance organ preservation. The authors provide **solid** evidence that an existing drug, SNC80, can rapidly and reversibly slow biochemical and metabolic activities while preserving cell and tissue viability. This study will be of interest to a broad set of readers interested in organ transplantation, tissue engineering, regenerative medicine organoids, and organ-on-a-chip engineering.

Tissue and organ loss to trauma, disease, and physical injury account for a large proportion of human ailments and approximately \$400 billion in annual medical burden.¹ The rapid and reversible slowing of metabolic and other physiological processes, here referred to as 'biostasis', can improve the survival of cells and organs for transplantation. This is currently accomplished clinically by lowering temperature and static cold storage is the standard of care for organ and tissue preservation; however, its long-term use can cause damage to the integrity of the graft.^{1,3} Hypothermia also has been used to induce a state of biostasis clinically using *ex-vivo* machine perfusion technologies, for example, during cardiac transplant surgery.

Combination of protective agents with perfusion and/or partial freezing approaches have extended preservation times in rat livers and whole pigs,^{4,5} and neural modulation approaches have been successful at demonstrating central control of body temperature and general metabolic state.^{6,7} However, both mechanical cooling and neural stimulation approaches are challenging to implement in a trauma triage or resource-limited situations, which would be better addressed by pharmaceutical interventions.

Several molecular strategies have been proposed to achieve metabolic suppression for organ preservation, including modulating H₂S, AMPK, opioid receptors, microRNAs, HO-1, and Nrf2.⁸ H₂S exposure has been reported to induce a hypometabolic state (reduced metabolism and body temperature) in a rodent model, although this was later found to be mediated in part by a low baseline laboratory temperature.⁹ The synthetic opioid peptide DADLE also can induce hypometabolism and improve organ preservation via a delta opioid receptor (DOR) mechanism, but with varied results.^{10,11} Compounds that directly impact glycolysis and mitochondrial respiration also have been considered for modulating organ metabolism, but many are acutely toxic and not reversible, fail to protect tissues from ischemic injury, or lack drug-like properties. Specifically, reduced functionality of the mitochondrial electron transport chain system without the endogenous antioxidant and anti-inflammatory molecules released by hibernators can result in the release of reactive oxygen species and cause tissue damage.⁸

Despite these efforts, there remains a large unmet need for improved tissue and organ preservation approaches across multiple application areas. Thus, in the present study, we set out to identify small molecules that could slow metabolism and mimic states normally induced by hypothermia, hibernation, or torpor, which could be used for preservation of living cells, tissues, and organs *ex vivo*, and potentially *in vivo* as well. Specifically, we required this biostasis approach to be rapidly inducible (< 1 hour) and safely reversible, with major tissue functions returning to control levels within 24 hours.

Through survey of the literature, we identified drugs with unintended side effects, such as lowering body temperature, which suggested they might have broad effects on metabolism. Among these drugs was SNC80, a small molecule drug originally developed as a nonaddictive pain reliever that acts independently of the mu-opioid pathway.¹² This drug caught our attention because it has been shown to induce hypothermia and protect against the effects of spinal cord ischemia in rodents.^{12,13} To explore whether SNC80 had potential to broadly slow tissue

metabolism and physiology, we administered the drug to *Xenopus laevis* embryos and tadpoles, which we and others have previously shown to be useful models for drug screening due to their small size, extrauterine development, and skin permeability to small molecules.^{14–17} The *Xenopus* tadpoles were dosed at higher levels than dictated by SNC80's defined IC₅₀ value, with the goal of exploring the effects of SNC80 outside its known pain relief properties and accentuating the drug's "off-target" effects, including hypometabolism.

Demonstration of physiological slowing in *Xenopus*

We first assessed SNC80's impacts on mobility, metabolism, and heart activity in *Xenopus*. Behavioral assessment in mobile tadpoles showed that 100 μ M SNC80 slows movement relative to vehicle-treated counterparts, reducing activity by approximately 50% within 1 hour, which is rapidly reversible when the drug is removed (**Figure 1a**). SNC80 also suppressed the rate of oxygen consumption to one-third of baseline within 3 hours of treatment (**Figure 1b**) and had extreme suppressive, but fully reversible, effects on heart rate after only 1 hour of exposure (**Figure 1c**). Oxygen consumption was reduced in immobile *Xenopus* embryos treated with SNC80 as well (**Figure S1a**), suggesting that its metabolism-suppressing effects are independent of the slowed movement observed in mobile tadpoles.

Using *in situ* Matrix-Assisted-Laser Desorption/Ionization-Time of Flight Mass Spectrometry Imaging (MALDI-ToF MSI), we assessed and visualized the biodistribution of SNC80 in the *Xenopus* tadpole. SNC80 was detected within 1 hour of treatment and it appeared in a punctate distribution in the gastrointestinal (GI) tract, gill region and skeletal muscle (**Figures 2a,b**), suggesting full-body delivery of the drug. Past lipidomic studies of hibernating mammals demonstrated the importance of lipid molecules for thermal adaptation during hibernation with significant alterations of lipidomic profiles being observed in liver, plasma, brain, skeletal muscle and cardiac muscle during torpor as compared to the animal's active state.^{18,19} Similarly, MALDI-ToF MSI analysis revealed significantly higher levels of acylcarnitine and cholesterol ester in SNC80-treated groups in both skeletal muscle and brain (**Figure 2c**).

Increased levels of long chain acylcarnitine associated with enhanced mitochondrial fatty acid oxidation are observed during fasting as well as in hibernating brown bears,^{20,21} and cholesterol ester levels increase in hibernating ground squirrels.²² While the increased acylcarnitine and cholesterol ester levels in brain suggest a shift towards beta oxidation of fatty acids, they also have been implicated in other pathways including antioxidant activity and neurotransmission.^{23,24}

Having confirmed that SNC80 induces slowing of multiple physiological parameters in the whole *Xenopus* organism, we then set out to explore the molecular basis for these effects. As SNC80 is a known DOR agonist,^{12,13} we next evaluated whether its effects could be mediated by delta opioids. However, when we tested the DOR antagonist, naltrindole (NTI), at double the concentration (200 μ M), it did not block the hypometabolic effects of SNC80 (**Figure 2d**), suggesting that SNC80 modulates metabolism independent of DOR activation.

Attenuating the delta opioid activity of SNC80

To further test if the physiological slowing effects of SNC80 are independent of DOR activation, we used a medicinal chemistry approach to explore whether we could separate its metabolism slowing and DOR activities. Toward this end, we designed molecular entities void of the distal basic nitrogen (N-ally group within SNC80), which we hypothesized was a critical DOR pharmacophore. This effort led to the development of a novel morpholino compound, WB3

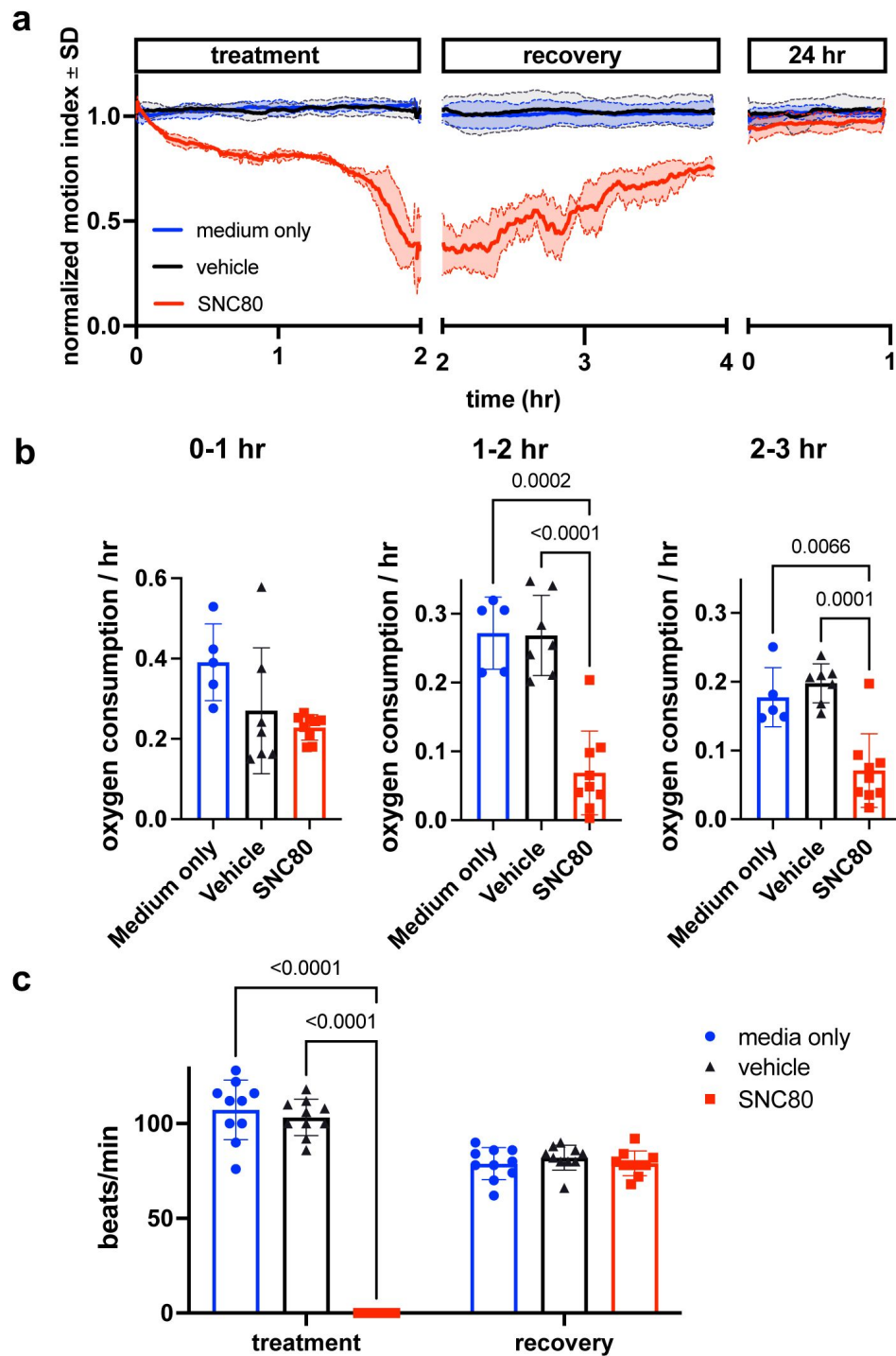


Figure 1.

SNC80 treatment in *Xenopus* tadpoles.

a. Swimming activity of *Xenopus* during 100 μ M SNC80 or vehicle treatment. Data represent the mean $\pm$ SD of $n=3$ replicates, with 10 tadpoles per replicate. **b.** Oxygen consumption rate for *Xenopus* treated with SNC80 or vehicle controls. $N=5$ medium only, $n=7$ vehicle and $n=9$ SNC80, with each data point representing the cumulative oxygen consumption for 5 tadpoles. **c.** Heart rate in SNC80 and vehicle-treated *Xenopus* ($n=10$ tadpoles/group). Statistical comparisons were performed using a Brown-Forsythe and Welch ANOVA test with a Dunnett correction for multiple comparisons (**b**) and a 2-way ANOVA (treatment $\times$ timepoint) with Tukey correction for multiple comparisons (**c**). Bar plots show the mean $\pm$ SD of each group.

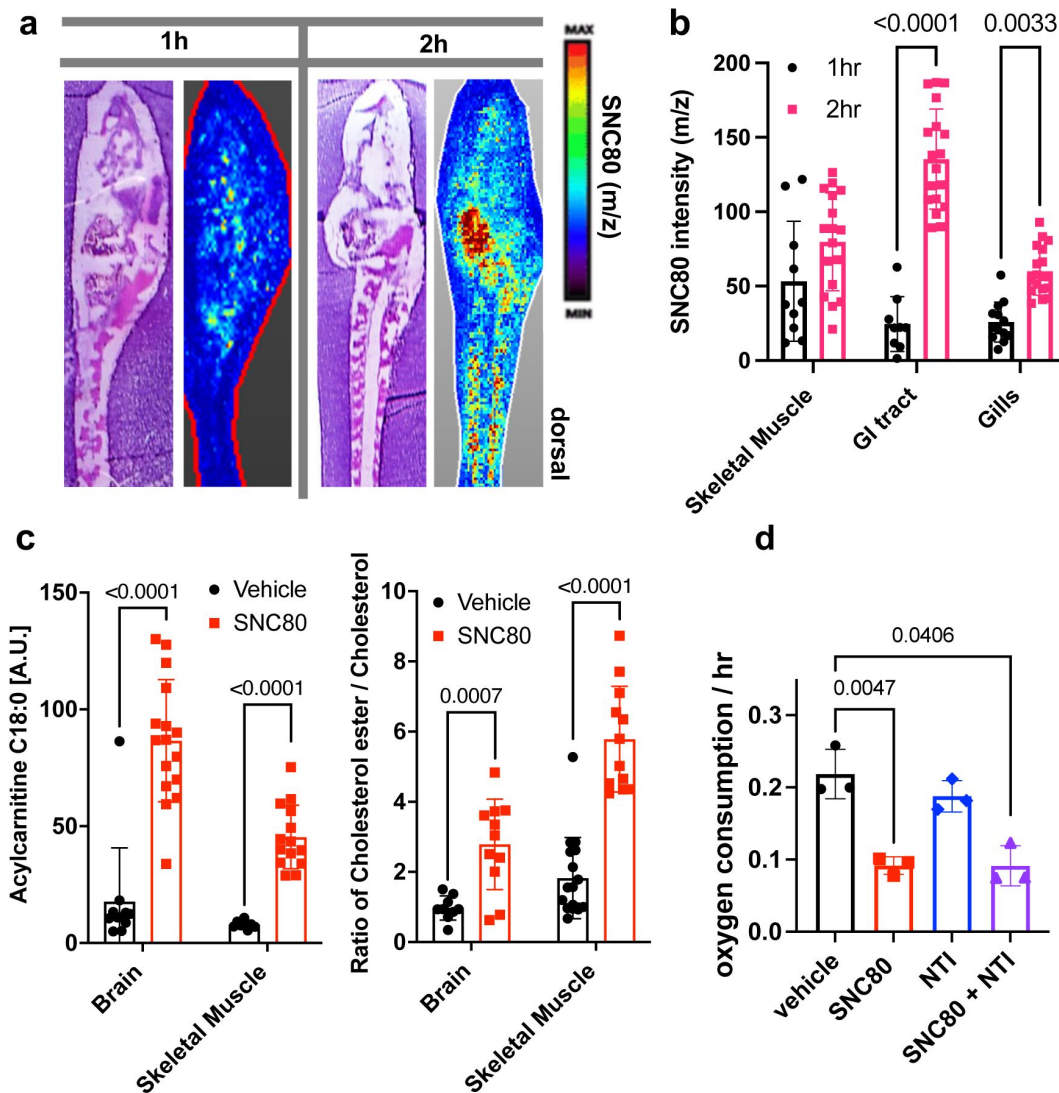


Figure 2.

SNC80 uptake and activity at the delta opioid receptor.

a. Distribution of SNC80 in *Xenopus* tadpoles after 1 and 2 hours of compound exposure. **b.** Uptake of SNC80 in the tadpole skeletal muscle, GI tract, and gills at 1 and 2 hours of exposure. **c.** Levels of acylcarnitine and cholesterol ester in the skeletal muscle and brain after 1 hour of SNC80 treatment. *In vivo* distributions of SNC80 and lipid levels measured from N=5 tadpoles per condition; N=3 sections/slide. **d.** Oxygen consumption in *Xenopus* tadpoles treated with SNC80, the delta opioid antagonist naltrindole, or a combination of SNC80 and naltrindole. N=3 replicates/group with each data point representing the cumulative oxygen consumption from 5 tadpoles. Statistical comparisons were performed using a two-way ANOVA (time x tissue region) with Sidak's correction for multiple comparisons (**b**), multiple unpaired t-tests for each tissue region with FDR correction (**c**), and a Welch ANOVA test with Dunnett correction for multiple comparisons between vehicle and each treatment group (**d**). Bar plots show the mean±SD of each group.

(**Figure 3a** [↗](#) & **Supplemental Synthesis Description**), which was synthesized and subsequently tested for DOR binding activity using a radioligand binding assay and non-linear fitting for specific binding to calculate the Hill Slope and IC₅₀ value. A specific binding curve could not be fit for SNC80 due to its high affinity and therefore its IC₅₀ is <4.6 nM, whereas WB3 exhibited an IC₅₀ of 3,470 nM (**Figure 3b** [↗](#)), suggesting almost 1,000 times less DOR binding activity for WB3 compared to SNC80. Subsequently, we screened WB3 in the *Xenopus* swimming and heart rate assays and found that the analog retains physiological slowing properties *in vivo* despite lacking delta opioid activity (**Figure 3c,d** [↗](#)). In fact, we observed more potent and rapid effects on swimming with exposure to WB3 compared to SNC80, suggesting that the non-opioid properties of SNC80 may be primarily responsible for its observed physiological effects.

Enhanced preservation of whole pig hearts and limbs perfused *ex vivo*

Based on these intriguing findings with human Organ Chips and *Xenopus* tadpoles, we set out to explore the potential clinical value of SNC80. Cardiac transplant is the gold standard for treating end-stage heart failure; however, it is hindered by donor shortage, limited cardiac graft preservation time, and lack of optimal organ preservation conditions. Currently, hearts are allocated to patients located within 4 hours of organ removal from the donor. Thus, to explore the potential utility of biostasis induction to lengthen the time between organ donation and recipient surgery through pharmacological induction of a hypometabolic state, we perfused surgically explanted whole porcine hearts with SNC80 using a portable, oxygenated perfusion preservation device (VPS ENCORE[®]) at subnormothermic temperatures (20–23°C). Prior work has demonstrated that the VPS ENCORE[®] enhances cardiac viability over standard of care-static cold storage and hearts perfused in this device for 4 hours demonstrate similar cardiac function to a healthy heart in both preclinical animal models and human deceased donor hearts.²⁵ [↗](#),²⁶ [↗](#)

We investigated the application of a biostasis inducer for preservation of the heart during 6 hours of perfusion in the VPS ENCORE[®] device. Importantly, with SNC80 treatment, we observed a rapid decline in the heart's oxygen consumption to < 50% of that observed in hearts exposed to the vehicle control, which was sustained over the 6-hour preservation period (**Figure 4a** [↗](#)). After preservation, hearts were removed from the VPS ENCORE[®] and placed in a Langendorff system for reperfusion and evaluation of oxygen consumption and cardiac contractility. This system assesses the left ventricular dP/dT without preload and afterload, thereby allowing the evaluation of the intrinsic contractility of the left ventricle. After 30 minutes of reperfusion and before defibrillation, SNC80-treated hearts had a lower oxygen consumption level than vehicle treated hearts (2.39 ± 1.23 versus 4.28 ± 2.48 ml/min/100g, respectively, at the 30-minute timepoint) (**Figure 4b** [↗](#)) that recovered to normal levels after defibrillation and with increases in temperature (2.49 ± 0.53 and 2.35 ± 0.31 ml/min/100g, respectively, at 60 minutes post-defibrillation) (**Figure 4c** [↗](#)). No evidence of substantial change in left ventricle (LV) contractility or relaxation were detected in SNC80-exposed hearts, either before or after they were defibrillated and exposed to epinephrine (**Figure 4d** [↗](#)).

Gene expression analysis of heart biopsy samples showed significant reduction of markers of inflammation (IL-6, IL-8, TNFα), hypoxia (Hif1α), and cell death (TP53, BCL2) in SNC80-treated hearts when compared to vehicle control hearts (**Figure 4e** [↗](#)), and no changes in the mitochondrial ATP synthase marker, ATP5MC3, were observed. Histopathological evaluation showed retention of normal muscle morphology (**Figure 4f** [↗](#)) without signs of tissue edema immediately after perfusion in the VPS ENCORE[®]. In some hearts treated with SNC80 greater waviness of muscle fibers was observed, possibly indicating a state of muscle contraction.

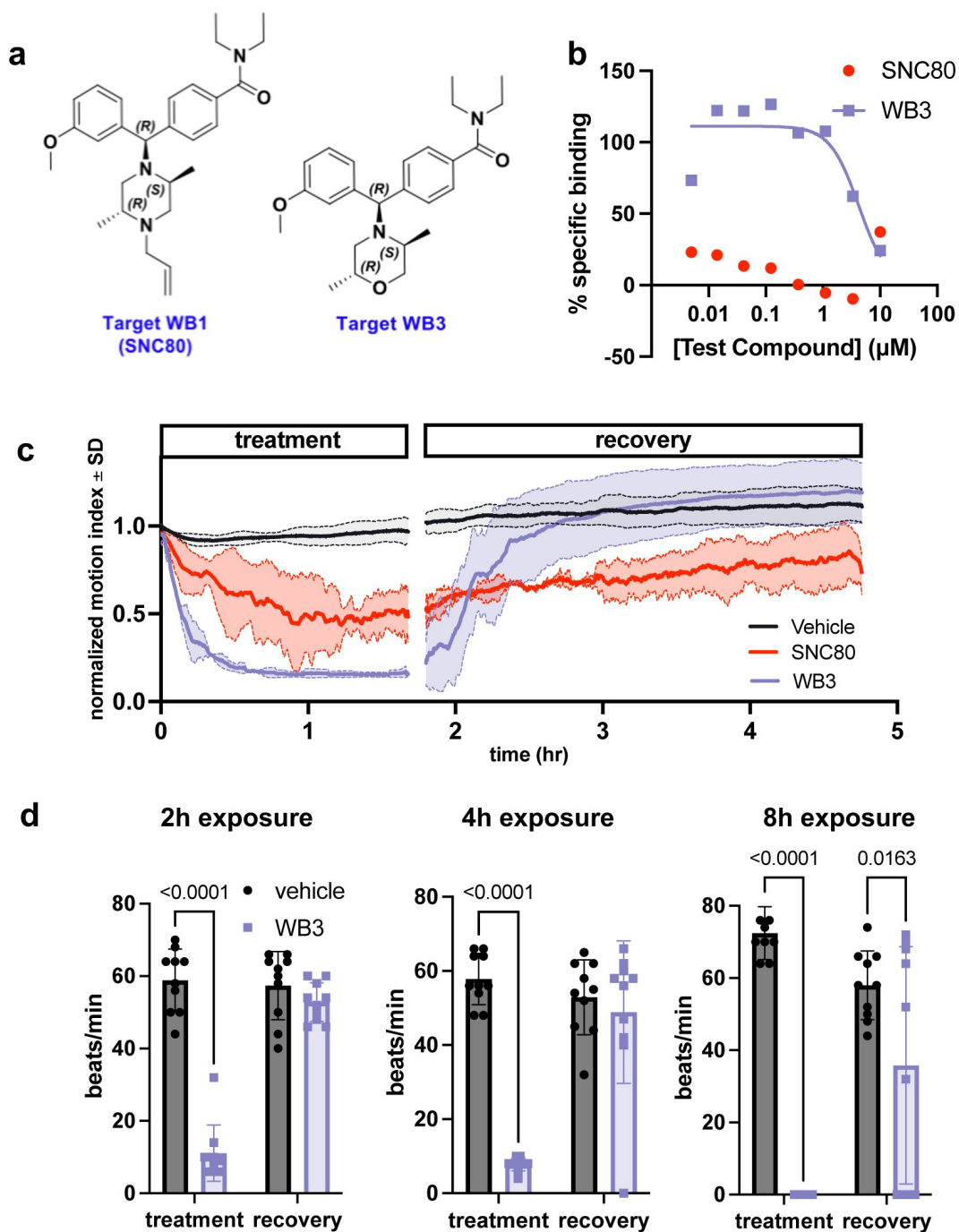


Figure 3.

Design and *in vivo* screening of novel analog WB3.

a. Molecular structures for SNC80 and novel compound WB3. **b.** Percent specific binding of the delta opioid receptor radioligand [3H]-DADLE in the presence of SNC80 and WB3. **c.** Swimming activity of *Xenopus* during 100μM WB3, SNC80, or vehicle treatment. Data represent the mean±SD of n=2 replicates, with 10 tadpoles per replicate. **d.** Heart rate in WB3 and vehicle-treated *Xenopus* (n=10 tadpoles/group). Statistical comparisons were performed using a two-way ANOVA (group x timepoint) with Sidak correction for multiple comparisons (**d**). Bar plots show the mean±SD of each group.

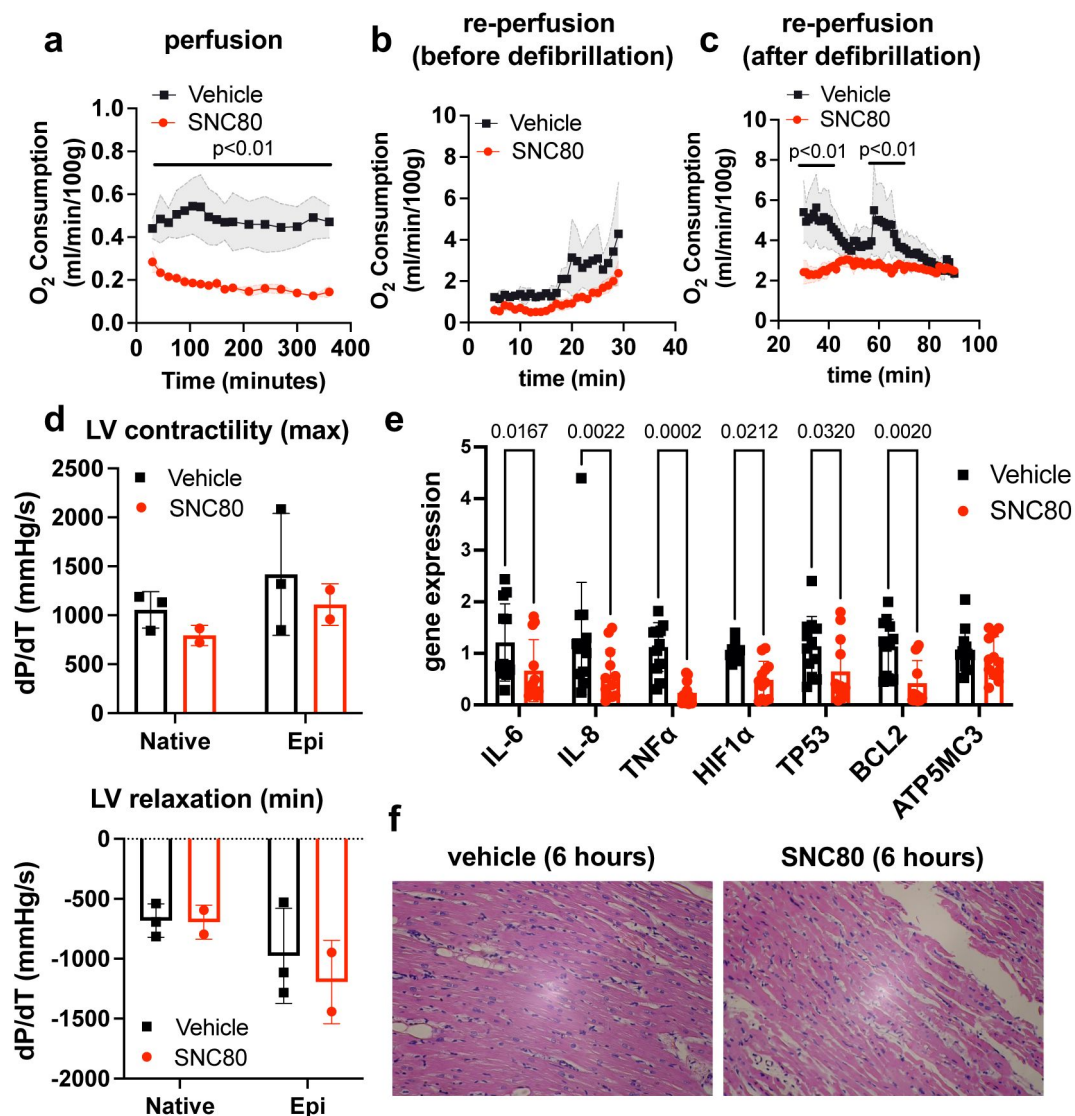


Figure 4.

Preservation of porcine cardiac grafts with SNC80 over 6 hours.

a. Oxygen consumption for vehicle and SNC80-treated cardiac grafts perfused in the VP.S ENCORE[®] device (n=3 hearts/group). **b.** Oxygen consumption during cardiac reperfusion on the Langendorff system before defibrillation (n=3 vehicle and n=3 SNC80 hearts). **c.** Oxygen consumption measured on the Langendorff system after defibrillation (n=3 vehicle and n=3 SNC80 hearts). **d.** Left ventricle (LV) contractility and relaxation before and after defibrillation and tissue exposure to epinephrine (n=3 vehicle and n=2 SNC-treated hearts). **e.** Gene expression analysis of heart biopsy samples for vehicle and SNC80-treated cardiac grafts (n=2 hearts/group; 3 technical replicates each). **f.** Light microscopic views of H&E-stained histological sections of heart biopsies (20x magnification) after 6 hours of perfusion. Statistical comparisons were performed using multiple unpaired t-tests between SNC80 and vehicle-treated cardiac grafts (**a,b,c**), two-way ANOVA with Sidak correction for multiple comparisons between SNC80 and vehicle-treated cardiac grafts (**d**), and unpaired t-tests between SNC80 and vehicle for each gene of interest (**e**). Data represent the mean $\pm$ SD.

Electrocardiogram (ECG) data revealed that both vehicle and SNC80-treated hearts exhibited irregular beating after perfusion and prior to defibrillation. After defibrillation with epinephrine, the P and QRS waveforms were visible in ECGs from 3 of 4 SNC80-treated hearts (**Table S1** [↗](#)), suggesting that those hearts regain atrial and ventricular polarization. The pulse rates of SNC80-treated hearts were at or near normal range for porcine hearts (70-120 beats/min) after defibrillation. Vehicle-treated hearts exhibited tachycardia following defibrillation, with all exhibiting pulse rates above the normal range for porcine hearts.

We also tested organ preservation in porcine limbs, where prolonged ischemia after amputation rapidly decreases the viability of skeletal muscle. Current preservation methods limit the success of limb reimplantation due to complications from rhabdomyolysis, necrosis, and reperfusion injury.²⁷ [↗](#) Again, we found that the relative metabolic rate measured during 3 hours of SNC80 exposure was significantly lower compared to vehicle-treated limbs (**Figure S2a** [↗](#)). Histopathological evaluation also showed retention of normal muscle morphology with no apparent changes in muscle bundle area (**Figure S2b** [↗](#)) or glycogen content (**Figure S2c** [↗](#)) suggesting that the tissue's potential for ATP synthesis was not compromised after SNC80 administration, although an increase in intercellular spacing was observed (**Figure S2d** [↗](#)). We did not observe full recovery of O₂ uptake at the 3-hour time point analyzed in this study (**Figure S2c** [↗](#)); however, the porcine limb contains a large amount of fat tissue that may be acting as a depot for the highly lipophilic SNC80, thereby leading to extended stasis under these conditions.

Elucidating potential SNC80 targets and pathways

To gain insight into the molecular basis of SNC80's hypometabolic and other physiological effects, we used thermal proteome profiling (TPP) to assess drug-protein binding and thereby identify previously unknown molecular targets of SNC80 that might be critical for its metabolism slowing activity. TPP of SNC80-treated *Xenopus* tissues revealed 19 direct and indirect binding targets (**Table 1** [↗](#)), including proteins involved in transmembrane transport, mitochondrion activity, and metabolic processes. Two potentially interesting binding targets of SNC80 are the excitatory amino acid transporter 1 (EAAT1; also known as SLC1A3) and the Na⁺/Ca²⁺ exchanger 1 (NCX1; also known as SLC8A1), both of which have been identified in cardiac tissue.^{28,29} [↗](#) In addition to their individual functions, mitochondrial ATP production has been previously reported to be stimulated through intermolecular interactions between EAATs and NCXs in neurons.³⁰ [↗](#) If this complex is also present in cardiac tissue, inhibition of EAAT and NCX1 could reduce cellular ATP levels,³¹ [↗](#) and thereby play a role in slowing metabolism.

Specifically, the blockade of Na⁺/Ca²⁺ exchange via interference with EAAT1-NCX1 interactions could prevent the influx of calcium into the mitochondria that is necessary for calcium-sensitive mitochondrial dehydrogenases that fuel ATP synthesis.³² [↗](#) However, the mitochondrial variant of NCX1 (NCLX) was not included in the *Xenopus* protein library used for TPP analysis so could not be fully evaluated in this study. TPP also identified two additional SNC80-binding proteins located within the mitochondrion: TOMM40 and COX6C. TOMM40 is a translocase located in the outer membrane of the mitochondria that has been implicated in tissue protective effects against late-onset Alzheimer disease³³ [↗](#) and COX6C is the terminal enzyme of the mitochondrial respiratory chain, catalyzing the electron transfer from reduced cytochrome c to oxygen.

In addition to interaction with mitochondrial proteins, we detected binding between SNC80 and enzymes involved in other metabolic processes, including alpha-L-fucosidase 2 (FUCA2), cysteine conjugate beta-lyase (CCBL1), retinoid isomerohydrolase (RPE65), and acyl-CoA Oxidase 3 (ACOX3). SNC80 also binds to multiple membrane-spanning proteins, including vacuole membrane protein 1 (VMP1) and ABCB1, an ATP-dependent translocase, in addition to NCX1 and EAAT1. The drug also bound to two proteins of the immune system, histocompatibility minor 13 (HM13) and pentraxin 3 (PTX3), as well as two cytoskeletal proteins, calmodulin regulated spectrin associated protein family member 2 (CAMSAP2) and integrin subunit alpha 7 (ITGA7). Interestingly, SNC80 also bound to four ribosomal proteins, RPL4, RPL27A, RPS25, and RRP9, which are frequent targets of

direct interaction	function	indirect interaction	function
CCBL1	- pyridoxal phosphate binding - cysteine-S-conjugate beta-lyase activity - L-glutamine aminotransferase activity - cellular modified amino acid metabolic process	ABCB1	- ABC-type bile acid transporter activity - regulation of fatty acid beta-oxidation - efflux transmembrane transporter activity
EAAT1	L-glutamate transmembrane transporter activity	ACOX3	- fatty acid beta-oxidation using acyl-CoA oxidase - fatty acid binding
FUCA2	- alpha-L-fucosidase activity - glycoside catabolic process	CAMSAP2	- cytoplasmic microtubule organization - calmodulin binding
HM13	- peptidase activity - aspartic endopeptidase activity	COX6C	- mitochondrial electron transport, cytochrome c to oxygen - generation of precursor metabolites and energy
PTX3	- negative regulation of glycoprotein metabolic process - complement component C1q complex binding	ITGA7	- integrin-mediated signaling pathway - cell adhesion mediated by integrin
RPE65	- phosphatidylserine binding - retinoid metabolic process	NCX1	- calcium:sodium antiporter activity - calcium ion binding - regulation of the force of heart contraction
RPL27A	structural constituent of ribosome cytoplasmic translation	ROBO1	- Roundabout signaling pathway - aorta development
RPL4	structural constituent of ribosome cytoplasmic translation	RRP9	- RNA binding - rRNA processing
RPS25	structural constituent of ribosome cytoplasmic translation		
TOMM40	- protein transmembrane transporter activity - protein targeting to mitochondrion - mitochondrial outer membrane translocase complex		
VMP1	- autophagosome membrane docking - plasma membrane phospholipid scrambling - lipoprotein transport		

Table 1.

SNC80 protein binding targets identified by thermal proteome profiling.

antibiotic and antiviral drugs. Collectively, these data suggest that induction of physiological slowing by SNC80 may involve simultaneous modulation of multiple pathways, with critical mechanisms involving mitochondria, membrane-spanning transporters, and possibly other metabolic processes as well.

Metabolic suppression in human Organ Chips

To explore whether this drug also induces a state of biostasis in human cells and tissues, as well as non-cardiac cells, we tested SNC80 in human organ-on-a-chip (Organ Chip) microfluidic culture devices that reconstitute tissue-tissue interfaces and human organ-level physiology with high fidelity *in vitro*.³⁴ The Organ Chip models reconstitute a tissue-tissue interface between organ-specific epithelium and endothelium across a porous extracellular matrix-coated membrane within a two-channel microfluidic device, and hence allow tissue-tissue cross talk observed *in vivo* that is not normally present in studies with cultured cells, as previously described.^{35,36} In this study, we used a Gut Chip lined by Caco-2 intestinal epithelial cells interfaced with HUVECs and a Liver Chip containing Huh7 liver epithelial cells interfaced with LSECs, respectively. However, we modified the Organ Chip design by integrating sensors that allow for live monitoring of oxygen (O₂) and transepithelial electrical resistance (TEER) (tissue barrier function) (**Figure S3a**). Both Organ Chips were treated with SNC80 by flowing the drug in medium for 2 days through the apical and basal channels of the chip, and then removing the drug followed by 7 days of additional culture. We confirmed that the intestinal epithelial cells underwent villus differentiation and accumulated mucus under these culture conditions (**Figure S3b**), in addition to maintaining a low oxygen environment after 10-16 days of culture (**Figure S3c**) as previously described.^{35,36} The Liver Chips also produced albumin and urea at levels (5 and 135 µg/day/chip, respectively) (**Figure S3d,e**) that were similar to those measured in Organ Chips lined by primary hepatocytes.^{37,38}

These studies revealed that SNC80 treatment reduced the respiratory (oxygen utilization) rate in the Gut Chip, resulting in an almost 6-fold increase in extracellular oxygen levels after 48 hours, which then progressively returned to baseline levels during the 7-day washout period (**Figure S4a**). Similar results were obtained with the more highly metabolic Liver Chip with SNC80 inducing higher levels of extracellular oxygen compared to control chips and this again reversed after drug washout (**Figure S4a**). Importantly, the tissues in both the human Organ Chips remained healthy after 2 days treatment with SNC80 as there was no significant change in barrier function (**Figures S5a,b**), cell growth (**Figure S3**), or cytotoxicity (**Figures S5c,d**). Similar to the data observed in cell cultures, ATP/ADP ratio in Liver Chips was significantly decreased after 24 and 48h of SNC80 treatment compared to the vehicle control chips. However, ATP/ADP ratio measurements between SNC80 and vehicle control were not significantly different after washout of the drug, indicating that Liver Chip tissues recover their normal metabolic function up to 4 days after treatment (**Figure S4b**).

Collectively, these data suggest that treatment with SNC80 in these physiologically relevant human cells and tissues cultured on-chip in an organ-relevant context resulted in a drop in respiration, with no switch towards glycolysis or reductive carboxylation, since the levels of lactate (**Figure S5e,f**) and glutamate remained unchanged in both chips (**Figure S5g,h**). Thus, in this organ-like microenvironment, the tissues decrease their total ATP production in presence of SNC80, which is accompanied by a global slowing of metabolism.

Discussion

Here we report the discovery that the small molecule drug, SNC80, and its analog WB3, are capable of slowing of multiple physiological processes, and thereby inducing a hypometabolic state without external cooling. Although there is extensive clinical experience using hypothermia to cryopreserve organs during surgery (e.g., cardiac bypass) and for donor transplants, comparatively little work has been done in the area of molecular induction of biostasis without external cooling. We found that SNC80 reversibly suppressed metabolism across multiple cell and tissue types, as well as different species and experimental models, at their basal temperatures, suggesting that this approach could be appropriate for preservation of a variety of cells, tissues, and organs. Critically, integrating this small molecule drug in a commercial organ preservation system with whole explanted pig hearts demonstrated potential for near-term translation and impact of this approach for applications relating to organ preservation and surgical transplantation.

Initially developed as a nonaddictive pain reliever, SNC80 was previously shown to induce hypothermia and protect against the effects of spinal cord ischemia in rodents,^{12,13} but its broad effects on metabolism were not known. In the present study, treatment with SNC80 slowed swimming movement and reduced oxygen consumption rates in a reversible manner in *Xenopus* tadpoles, and it lowered the metabolic rate in non-mobile *Xenopus* embryos as well. Importantly, SNC80 displayed similar metabolism suppressing activities that were reversible in physiologically relevant human Organ Chip microfluidic devices that reconstitute organ-level structures and functions in vitro. These data suggested that induction of a hypometabolic state by this small molecular drug could potentially provide a new paradigm to prolong organ preservation. Indeed, when we perfused explanted porcine hearts and limbs with SNC80, we experimentally confirmed that this biostasis agent can slow metabolism in a whole complex living organs, providing direct support for its potential integration into commercial organ preservation devices that strive to extend the viability of human organs for surgical transplantation.

SNC80 is defined as a highly selective non-peptide DOR agonist drug and other DOR agonists, such as the enkephalin DADLE, have been shown to reduce metabolic demand and protect against oxidative stress, thereby decreasing injury and apoptosis in perfused organs.^{8,10,11} Prior studies of DADLE found that the DOR pathway was required to alter metabolism and provide protective effects, however, we found that SNC80 suppresses metabolism via a mechanism that is independent of its DOR modulating activity. Instead, we found that SNC80 interacts with multiple proteins involved in transmembrane transport, mitochondrion activity, and metabolic processes. Interestingly, one of the identified targets, NCX1, has been previously implicated in control of temperature compensation and autonomous oscillation,³⁹ suggesting that it plays a key role in the regulation of biochemical reaction speed, cell cycle, and other forms of biological timing, such as circadian rhythm. In addition to its effects on tissues of the nervous system, multiple drugs that target $\text{Na}^+/\text{Ca}^{2+}$ exchange by NCX, such as the KBR family of drugs, were developed for the treatment of ischemia, reperfusion injury, and heart arrhythmias.⁴⁰ In rats, pre-ischemic or post-ischemic treatments with KBR compounds improved recovery of ventricular pump function and reduced reoxygenation-induced injury as measured the reduced release of intracellular components (e.g., lactate dehydrogenase).^{40,41} Our results suggest that SNC80 could be acting on the NCX1 transmembrane protein and/or NCX1/EAAT1 pathway may result at least in part from induction of a hypometabolic state.⁴⁵ Of the tissues impacted by SNC80, the observed effects were most profound on *Xenopus* tadpole heart function and porcine hearts. SNC80 and other opioids have been shown to decrease cardiac inotropy and contractility,^{42,43} and prior work has demonstrated that decreases in cardiac inotropy and contractility by opioid agonists may be mediated by opioid receptor-independent actions of these drugs.⁴³ Substantial slowing of heart rate by WB3, which exhibits approximately 1,000 times less delta opioid activity than SNC80, further supports this claim. Similar to our findings by TPP, other investigators identified non-opioid receptor-mediated actions of these compounds on ion channel function.⁴³ Changes in

contractility were not observed immediately after perfusion with SNC80 in porcine hearts tested here, likely because the SNC80 was washed out of the tissue before contractility was measured; however, future work should continue to monitor these parameters, especially at later time points. Although bradycardia likely contributes to the hypometabolism observed in both systems, we also see reduced metabolic rates in systems lacking cardiac cells, and thus the hypometabolic activity of SNC80 we observed is not completely cardiac-specific.

The *Xenopus* and perfused porcine heart systems may be particularly amenable to extreme slowing of cardiac output because these live tissues are surrounded by well-oxygenated medium during treatment, potentially avoiding tissue injury sometimes associated with hypometabolic states. However, the assumption that high oxygen content in the perfusate completely precludes ischemia may not be valid. Assuming adequate flushing during recovery, local vasospasm or a fat micro embolism occurring during perfusion could restrict perfusion flow thereby inducing a localized hypoxic state and HIF1 α expression in the tissue. SNC80 may be acting in a vasodilatory capacity, thereby facilitating oxygen delivery and suppressing HIF1 α expression. In addition, although HIF1 α is primarily known for its role in response to hypoxia, it has some oxygen-independent functions and can be induced by factors other than oxygen levels, such as growth factors and cytokines, which can activate its transcriptional activity even under normoxic conditions. In these cases, HIF1 α can regulate genes involved in cell proliferation, metabolism, inflammation, and other cellular processes. Therefore, observation of suppressed HIF1 α , as well as the reduction of multiple inflammatory and cell death markers, in the treated porcine cardiac tissue suggests a protective effect of SNC80.

Before considering broader applications of SNC80 or any of its analogs, such as whole-body infusion during trauma or infection, it will be critical to consider impacts of this drug not only on the heart but other organ systems as well. In particular, our *Xenopus* data suggest that there could be substantial uptake of SNC80 in the brain, thus potentially impacting tissues of the central nervous system. Therefore, future work should characterize cognitive recovery in organisms receiving systemic delivery of stasis inducers. In addition, whole-body infusion applications may require further pharmaceutical development, including chemical modifications to SNC80 and development of targeted delivery methods to ensure clinical safety.

In summary, our findings demonstrate that the small molecule drugs SNC80 and WB3 can be used as a biostasis inducer to produce a hypometabolic state, which can provide potential therapeutic value for clinical applications, such as enhanced preservation of organs. The use of a biostatic compound in combination with a self-contained transport system could potentially increase organ viability for sustained times *ex vivo*, thus helping to advance organ and tissue preservation for surgical transplantation. The ability to induce a reversible hypometabolic, suspended animation-type state rapidly through injection at the point-of-care also could enable new therapeutic approaches to slow the detrimental effects of trauma and acute infection, thereby increasing patient survival in military settings and space exploration, as well as in civilian health emergency operations. It also may be possible to use this biostasis approach to slow metabolic activities in the brain in cases of stroke or traumatic brain injury, indications for which current treatments are severely limited.

Methods

Xenopus model

Xenopus embryos were fertilized at Tufts University using procedures reviewed and approved by the Tufts University Institutional Animal Care and Use Committee regulations and transferred to the Wyss Institute. *Xenopus* embryos and tadpoles were housed at 18°C with a 12/12 h light/dark

cycle in 0.1X Marc's Modified Ringer's (MMR) medium. All animal experiments and procedures were reviewed and approved by the Harvard Medical School (HMS) Institutional Animal Care and Use Committee regulations.

Drugs used in the *Xenopus* studies were dissolved in DMSO to a stock concentration of 10 mM and diluted to their final concentrations with a 1% DMSO concentration or less. Media with dosed compound were made fresh and screening was performed in tadpoles at Stages 46-50. The activity of free-swimming tadpoles during and after exposure to drugs were recorded in 60 mm dishes using a SONY Alpha a6100 camera with 16 mm objective (Sony Corporation, Tokyo, Japan) against an illuminated background. Mobility was quantified in Matlab (Mathworks, Natick, MA) by mapping differences between frames to a movement index (0 – 1), with 0 indicating no movement and 1 indicating maximal movement. Oxygen consumption was measured during *Xenopus* exposure to drugs as a proxy for whole-organism metabolism using the Firesting Optical Oxygen Meter and the Pyro Oxygen Logger software (version 3.317) (Pyroscience GmbH, Aachen, Germany). Oxygen consumption for *Xenopus* embryos was measured using the OxoPlate (PreSens Precision Sensing GmbH, Regensburg, Germany). Oxygen consumption rate for each sample was measured using a linear fit to the oxygen consumption curves in Prism (GraphPad, San Diego, CA). Heart rate was measured by microscopy imaging in *Xenopus* following 1, 2, 4, or 8 hours of drug exposure. Vehicle and drug-treated groups were briefly incubated in 0.01% tricaine (Syndel, Ferndale, WA) to minimize tadpole movement during heart imaging. Videos of heart function were acquired using a ZEISS Axio Zoom.V16 microscope and ZEN BLUE Microscopy software (version 3.1) (Carl Zeiss AG, Oberkochen, Germany) at a rate of 1 frame/100 ms. Heart beats were counted automatically using a custom code in Python. At the completion of experiments, *Xenopus* were euthanized by immersion in 0.2% tricaine for 30 minutes, followed by fixation or bleaching and disposal.

MALDI-ToF

Tadpoles were embedded in gelatin (0.11 g/mL) and stored in -80°C. Embedded tadpoles were sectioned into 16 µm-thick slices using a Leica CM1850 (Leica Microsystems, Wetzlar, Germany) onto indium tin oxide (ITO)-coated glass slides (Bruker Daltonics, Billerica, MA) and immediately placed in a vacuum desiccator before matrix application (N=5 tadpoles per condition; N=3 sections/slide). Experiments were duplicated to confirm the reproducibility. The matrix solution was composed of 2,4 Dihydroxybenzoic acid (40mg/mL) in 0.1% formic acid, methanol/water (1:1 v/v) and was made fresh for each application. The HTX-sprayer nebulizer (HTX Technologies, Carrboro, NC, USA) was used to apply the matrix at 80°C at 24 passes over the sample, at a flow rate of 50 µL/min, 10 psi pressure and a track speed at 1,250 mm/min. Before imaging, SNC80 spotting with matrix and *Xenopus* lysate was performed to determine the optimal matrix for the best S/N ratio and the least ion suppression from *Xenopus* tissue and to identify which adducts were detectable for each compound for imaging. *In situ* MALDI-ToF imaging was acquired using the rapifleX (Bruker Daltonics, Billerica, MA), in positive ion mode at 1000 Hz in the mass range of 300-990 m/z. The laser raster step spacing was set to 45 µm step size at 500 shots per pixel. External calibration was performed using a red phosphorus slurry drop casted on a region of the ITO slide without tissue. Preliminary processing of the data was performed using FlexAnalysis (Bruker Daltonics, Billerica, MA). Images were normalized by the total ion count and baseline-corrected using Top Hat algorithm with a _baseline width. After MALDI-ToF MSI acquisition, ITO slides were then stained for H&E and imaged using the Cytation 5 (Agilent, Santa Clara, CA) and Epsilon scanner at 3000 dpi to select regions of interest using the open-source program, Qupath.⁴⁴ Mass selection windows and signal intensity analysis was conducted with SCiLS Lab (Bruker Daltonics, Billerica, MA), where ions of interest, including SNC80, acylcarnitine and cholesterol ester C18, were chosen with a width of ±0.25 Da. Reference peaks of target analytes were based on previous MALDI-ToF work in *Xenopus* tissues from previous findings,⁴⁵ and during spotting of target analytes with MALDI-ToF MS/MS. The mean signal intensity of the defined regions of interest were obtained with each ion of interest.

Radioligand binding assay

To assess binding of WB3 to the delta opioid receptor, we tested WB3 and SNC80 at 8-concentrations with 3-fold serial dilutions starting at 10 μ M (Reaction Biology, Malvern, PA). Binding of the tracer [3H]-DADLE (10 nM) was assessed in the presence of each concentration of WB3 and SNC80 and the [3]-DADLE signal was fit to a dose-response curve to quantify the Hill Slope and IC50 values.

Thermal proteome profiling

To identify drug-bound targets based on altered thermal stability, *Xenopus* tadpoles were treated with either SNC80 or vehicle for 2 hours. *Xenopus* were euthanized in 20X tricaine and then transferred back to their respective drug solutions to ensure drug effect is not lost during heat treatment. *Xenopus* samples were incubated at 9 temperature points (30C, 35C, 40C, 44C, 48C, 52C, 56C, 60C, 65C) using a PCR machine (3 minutes heat time per condition, 3 tadpoles per condition) for both drug treated and vehicle treated groups. After transfer to room temperature, liquid was aspirated from samples and samples were snap frozen in liquid nitrogen. Samples were lysed in 100 μ L of PBS + 0.4 % NP-40 with 4 cycles of freeze/thaw. 11 μ L of each of the 9 temperatures were pooled in one tube and samples were centrifuged at 13,000 RPM for 75 minutes, at 4C. Proteins were reduced and alkylated with 10 mM DTT and 15 mM IAA for 15 minutes at 65C and 30 minutes at room temperature, respectively. Proteins were then precipitated with 8 volumes of ice-cold acetone and 1 volume of ice-cold methanol overnight. Digestion was carried for 4 hours in 50 mM Tris pH 8 + 0.75 mM Urea + 1 μ g trypsin/LysC at 37 Celcius with agitation. Another 1 μ g of trypsin/LysC was added and digestion was continued overnight. Peptides were purified by reversed phase SPE and analyzed by LC-MS/MS (PhenoSwitch Bioscience, Sherbrooke, QC, Canada). LC-MS/MS acquisition was performed with a ABSciex TripleTOF 6600 (ABSciex, Foster City, CA, USA) equipped with an electrospray interface with a 25 μ m iD capillary and coupled to an Eksigent μ UHPLC (Eksigent, Redwood City, CA, USA). Analyst TF 1.8 software was used to control the instrument and for data processing and acquisition. Samples were acquired twice in SWATH acquisition mode using gas phase fractionation. The source voltage was set to 5.5 kV and maintained at 325C, curtain gas was set at 45 psi, gas one at 25 psi and gas two at 25 psi. Separation was performed on a reversed phase Kinetex XB column 0.3 μ m i.d., 2.6 μ m particles, 150mm long (Phenomenex) which was maintained at 60C. Samples were injected by loop overfilling into a 5 μ L loop. For the 60 minutes LC gradient, the mobile phase consisted of the following solvent A (0.2% v/v formic acid and 3% DMSO v/v in water) and solvent B (0.2% v/v formic acid and 3% DMSO in EtOH) at a flow rate of 3 μ L/min. GPF files for both acquisitions were analyzed on a previously generated 3D ion library. Each GPF file was analyzed using 10 peptides per protein, 4 MS/MS transition per peptide, 12.5-minute RT window and 25 ppm XIC width in Peakview (Sciex). The reported quantification for a protein is the sum of all the correctly integrated peptides (FDR<0.05) in both GPF files.

Human Cell Cultures

Caco2 intestinal epithelial cells were grown in high glucose (4.5g/L) DMEM (Gibco), 10% FBS (Gibco) and 1% penicillin/ streptomycin (Gibco) and subcultured every 2-3 days at 1:6 ratio. For all experiments, cells were used between passage 59 and 64. HUVE cells were maintained in EBM-2 (Lonza) media with EGM-2 BulletKit (Lonza), 2% FBS and 1% penicillin/streptomycin (Gibco) and subcultured every 2-3 days at 1:6 ratio. For all experiments, cells were used between passage 3 and 7. HuH-7 liver epithelial cells were maintained in DMEM with 4.5g/L glucose (HyClone), 10% FBS and 1% penicillin/ streptomycin (Gibco). We also obtained wild-type Huh7 transduced with lentivirus particle pLV[Exp]-Puro-CMV>{Perceval(ns)};T2A:mCherry/3xNLS (Vector Builder) that were used for live imaging of changes in ATP/ADP ratio within these cells, which directly correlates to cellular metabolic activity.⁴⁶ LSEC cells were maintained in EBM-2 (Lonza) media with EGM-2 BulletKit (Lonza), 2% FBS and 1% penicillin/streptomycin (Gibco).

Metabolic assays

Effluents were collected at different timepoints (baseline, 24h, 48h and 1-3 days recovery after washout) throughout the exposure to SNC80 or 1:1000 DMSO vehicle. After treatment, the effluent was collected and analyzed for lactate and glutamate production using the Lactate-GloTM (Promega, #J5021) and Glutamate-GloTM (Promega, #J7021) assay kits respectively as per manufacturer protocols.

Cytotoxicity assay

Cytotoxicity measurements on the Organ Chip effluents were performed using the LDH-GloTM (Promega, #J2380) as per manufacturer protocols.

ATP/ADP ratio measurements

Huh7 cells were transfected with a reporter that enables direct measurements of cellular ATP:ADP ratio. The basal excitation spectrum of the reporter protein was similar to that of YFP (peak around 490 nm) but it had an additional peak at CFP wavelength (405 nm). As ATP levels decreased, more ADP can bind to the site, decreasing the YFP/CFP ratio value. BioTek Cytation 5 Cell Imager (Agilent) was used to take RFP, YFP and CFP images of each well every 20min for 180 minutes upon exposure to 100uM SNC80 or 1:100 DMSO vehicle. Post processing of the images was performed using the open source CellProfiler program.⁴⁷ We implemented the following pipeline to obtain quantitative ATP/ADP ratio values: (1) Identify nuclei with RFP marker, (2) Define a cell as 4pixels around the edge of the nuclei, (3) Measure cells YFP and CFP intensity and YFP/CFP ratio, (4) Count cells per Image, (5) Track cells YFP and CFP intensity over the time course (0-3h with 20min intervals), (6) Measure YFP/CFP ratio average **over all cells** per image and normalize to cell count (7).

For the qualitative images, raw images were processed with ImageJ by using the image calculator to divide yfp/cfp intensity of respective images. “Rainbow RGB” lookup table was used to show the intensity changes. Tones towards the red show more intensity, higher ATP/ADP ratio and higher metabolism.

Organ Chips with Integrated Sensors

To allow for live oxygen and transepithelial/endothelial resistance (TEER) monitoring, we used our in-house developed and fabricated microfluidic chips with integrated TEER and oxygen sensors. To mimic the in vivo-like cell-cell interface on the models, a dual-channel PDMS microfluidic chip was used where the apical channel is separated from the basal channel with a porous PDMS membrane. A pair of TEER sensing electrodes were integrated on the apical side of the top channel and the basal side of the bottom channel to measure the electrical resistance across the epithelium-endothelium tissue interface. The O₂ sensing nanoparticles (OXNANO, PyroScience) were integrated at four different locations in the chip: two in the apical channel and two in similar locations in the bottom channel. This is to allow a fair representation of the whole chip oxygen tension levels.

After fabrication, the chips were sterilized by plasma treatment and desiccated for 30 minutes. The polymeric surfaces of the membrane were then chemically functionalized using ER1/ER2 buffers following EmulateTM Inc. protocol and coated with extracellular matrix (ECM) for 1.5-2 hours at 37°C. Gut Chip apical and basal channels were coated with 30 µg/ml Collagen I (Advanced BioMatrix, Cat. no. 5005) and 100 µg/ml Matrigel (BD Biosciences) in cold PBS. Liver Chip apical channel was coated with 30 µg/mL fibronectin (Corning, Cat. no. 356008) and 400 µg/mL Collagen I (Advanced BioMatrix, Cat. no. 5005) in cold DPBS, while the basal channel was coated with 100 µg/mL Matrigel (BD Biosciences) and 200 µg/mL type I collagen in cold DPBS. To create Gut Chip, HUVECs (6-8×10⁶ cells/ml (P5)) were first seeded on the bottom side of the permeable membrane in the basal channel and incubated for 2-3 hours at 37°C before seeding the Caco-2 cells (2-3×10⁶

cells/ml (P78)) on the membrane in the apical channel. Liver Chip was similarly created by seeding the LSEC (4.5×10^6 cells/ml (P3-8)) the first in the basal channel followed by seeding the Huh7 hepatocytes ($2.5\text{--}3.5 \times 10^6$ cells/ml (P5-15)) on the membrane in the apical channel. The chips were then cultured under continuous flow using the commercial culture module instrument (ZOE™ Culture Module, Emulate Inc., USA). The Gut Chips were maintained under continuous flow of 30 μ l/hr in the apical and basal channels for 10 days before changing their respective media to differentiation-promoting media (5% FBS in DMED apically and 0.5% FBS endothelial medium basally) for the rest of the culture time. The Liver Chips were also maintained under continuous flow of growth media at 30 μ l/hr in the apical and basal channels for 5-6 days before reducing the FBS content to 5% in Huh7 media and 0.5% in LSEC media for inducing tissue differentiation.

Human Organ Chip studies

SNC80 was tested on the Gut Chip at day 16 of culture, when the epithelium showed characteristics of matured, functional tissue and the Liver Chip was tested after 7-10 days of culture. SNC80 [100 μ M] in the culture media was perfused through the apical and basal channels of the chips at 30 μ l/hr, while the control chips were perfused with the same culture media containing the compound's vehicle [0.5% DMSO] using the same flow condition. The chips were treated with SNC80 for 48 hours before the drug was removed from the media and the culture was continued for another 4 days. Both Liver and Gut Chips were monitored at d-1, 0h, 3h, 24, 48h timepoints as well as day 1 and 3 after the drug washout for oxygen consumption, TEER measurements, and effluent analysis. Fluorescence microscopy was continuously performed from 0h throughout the treatment period and up to 3 days of recovery. CellProfiler program was used to analyze ATP/ADP ratio as described above.

Oxygen and TEER measurements on-chip

Oxygen tension level measurements were performed on-chip during the culture using FireSting®-O2 Optical Oxygen and Temperature Meter (FSO2-C4, Pyroscience sensor technology, Germany) connected to Pryo Oxygen Logger (V3.317©, Germany). TEER measurements were performed on-chip using our in-house developed multiplexed TEER Tray and remotely measured from outside the incubator. A four-point impedance measurement was taken on-chip periodically every 24 hours during the SNC80 treatment and recovery using the IviumSoft application (V4.97, Informer Technologies, Inc.). To obtain the TEER of only the epithelial-endothelial tissue interface, the measured impedance at 100Hz was subtracted from the impedance at 100kHz and reported as the tissue impedance (Ω) throughout the culture and in response to different stimuli as we reported previously.⁴⁸

DNA damage and proliferation assays

The Click-IT™ EdU cell proliferation kit (ThermoFisher, #C10337) was used to assess cell proliferation in the formaldehyde-fixed chips and DNA damage was assessed by staining anti-H2AX antibody (Novus Bio NB100-2280).

Urea Assay

Urea nitrogen was measured using a colorimetric assay (urea nitrogen test, Stanbio Laboratory), whereby undiluted liver chip effluents were added to a 96 well plate and mixed with the kit's working reagents as per manufacturer protocol.

Albumin Assay

Liver albumin production was measured using an enzyme-linked immunosorbent assay (Bethyl Laboratories). Liver Chip effluents were diluted 1:100 in buffer before being plated in the kit's 96 well plate, aside from standard wells with dilution series of known albumin concentration.

solution. We followed manufacturer protocol to perform the sandwich ELISA assay and measure absorbance at 450nm.

Porcine heart ex vivo perfusion studies

After induction of anesthesia and preparation for clean, non-survival surgery, a median sternotomy was performed followed by longitudinal incision in the pericardium to expose the heart and the great vessels. Following blunt dissection and exposure of the vena cava, aorta and the brachycephalic artery, the superior vena cava, and azygos vein was ligated and divided. The brachycephalic artery was catheterized for infusion of cardioplegia solution. The inferior vena cava was clamped and divided proximal to the clamp. Two or three pulmonary arteries were incised for decompression of the left side of the heart. The aorta was immediately clamped and 10-15 ml/kg of body weight of cold (2-4°C) cardioplegia solution (del Nido without lidocaine, 3 L) was infused at a pressure of 70-80mmHg to stop the heart. Cold 4°C saline and ice were poured over the heart and the accumulated fluids from the thoracic cavity were removed by suction. Following cessation of myocardial function, the heart was removed from the chest by further dissection and division of the pulmonary vessels and aorta distal to the clamp. The heart was cooled in a surgical bowl with cold saline for inspection and preparation for perfusion. All procedures were conducted under the UT Health San Antonio approved Institutional Animal Care and Use Committee (IACUC) protocol 20210080AR.

After the heart was removed and placed in a surgical bowl, the aorta and the coronary sinus were cannulated using sterile zip ties. The heart was placed into the VPS ENCORE[®] perfusion device filled with 3 L of Krebs-Henseleit buffer supplemented with 20PEG and hemoglobin-based oxygen carrier (HBOC; Hemoglobin Oxygen Therapeutics, LLC, Souderton, PA). The heart was perfused with a pressure of approximately <20 mmHg resulting in a coronary perfusate flow of 50-100 mL/min at a 20-23°C temperature. Hearts were immediately treated with 3 mL of 3% lactic acid containing 135 mg of SNC80 (100 µM final concentration), administered through the arterial line followed by the 3 mL of Krebs-Henseleit flush. Post drug administration, the heart continued to be perfused for additional 6 hours.

Two oxygen probes (Pyro FireStingTM) collected oxygen partial pressure (mmHg) data of the solution before entering the coronary arteries (arterial) and after exiting the cannulated pulmonary artery (venous), whereas flow was measured using the Sensirion flow meter. Oxygen consumption was captured using the following formula, $(([O_2]_a - [O_2]_v) / 100 * Q) / \text{heart weight} * 100$. $[O_2]_a$ and $[O_2]_v$ are defined as oxygen content of arterial and venous perfusate respectively. $[O_2]$ was calculated as $(1.34 * Hb * SO_2) + (K * pO_2)$; where 1.34 is mL O₂ / Hb (g), Hb is the concentration of hemoglobin measured in g/dL, SO₂ is the oxygen saturation ratio calculated using the equation developed by Severinghaus,⁴⁹ K is the oxygen solubility coefficient adjusted for the perfusate temperature at every data point, pO₂ is the partial pressure of oxygen in mmHg for the perfusate sample, and Q is coronary flow in mL/min.

After preservation, hearts were placed on a Langendorff system with a perfusate mixture of 1:1 porcine blood and Krebs-Henseleit buffer. Myocardial oxygen consumption (MVO₂ mL O₂ / min / 100g) was calculated during an initial 30 min of rewarming stage (before defibrillation) and after defibrillation using two oxygen probes (Pyro FireStingTM, placed in arterial and venous effluent). Oxygen consumption was⁵³ calculated using the same approach described for the VPS ENCORE[®] perfusion stage. The left ventricular function was expressed as dP/dT (mmHg/sec) measured by placing a pressure catheter (Millar, 5F) in the left ventricle. Data was analyzed using PowerLab (LabChart 8.1.16) blood pressure analyzer. Data was selected from the left ventricular pressure waveform as the average of 30 beats after stabilization (approx. 30 minutes after defibrillation). Contractility after inotropic support was measured less than 5 minutes after administration of epinephrine (0.5mg injected arterial). End diastolic pressure of the left ventricle was maintained 5.14 +/- 1.89 mmHg through administering perfusate directly into the left ventricle through the pulmonary veins and venting to atmosphere.

ECG data were obtained through the placement of leads on the apex (positive), right atrial appendage (negative), and aortic root (ground). Data was collected through the PowerLab system and 30-second time intervals were selected for all groups before the flush, before defibrillation, at native contraction, and after epinephrine drug administration. The ECG amplitude (mV) was analyzed using LabChart Pro (v8.1.19) peak analysis detection at approximate time points 10min (before flush) and 30min (before defibrillation) to detect the average amplitude of electrical activity without full contraction. LabChart Pro (v8.1.19) ECG Analysis detection was used at time points 35min (Epinephrine) and 60min (Native) to detect the average amplitude of the QRS complex with contraction. The native QRS Interval (s) and ST Height (mV) was averaged over a 30-second time interval using LabChart Pro (v8.1.19) ECG Analysis.

Total RNA from cardiac tissue biopsies (2.5 mm) were extracted following the RNeasy Mini kit (Qiagen) according to the manufacturer's instructions. The concentration of total RNA was evaluated and measured at 260/280nm by spectrophotometer (NanoVue Plus, GE Healthcare). Synthesis of cDNA was performed from 0.5 µg of total RNA, which was reverse transcribed using (iScript cDNA Synthesis Kit, Bio-Rad) according to the manufacturer's instructions. Gene-specific pre-designed oligonucleotide primers were purchase from Sigma. qRT-PCR was done using SsoAdvanced Universal SYBR Green Supermix and SFX96 Touch real time PCR detection system (Bio-Rad, T100 Thermal Cycler). The cycling parameters were as follows: initial denaturation 95°C, 2 min; denaturation 95°C, 5 s; annealing/extension 60°C, 30 s; number of cycles 40; melt curve 65°-95°C (0.5°C increments). The comparative CT (2- $\Delta\Delta$ CT) method was used for all quantification. Values were normalized to the Gapdh housekeeping gene.

Porcine limb ex vivo perfusion studies

Hindlimbs (N=8) were procured via protocol tissue sharing at the Joint Base San Antonio Clinical Investigation and Research Support (CIRS) building, following IV Pentobarbital euthanasia, 100 mg/kg from a host protocol. Host protocol animals were treated with the same surgical manipulations and no chemicals or drugs were introduced experimentally prior to euthanasia. Following euthanasia, a curvilinear incision was made around the hindlimb circumferentially. Subsequently, the neurovascular pedicle in the medial hindlimb was identified and carefully dissected out to preserve the vascular bundle. The pedicle was then divided at the origin of the artery and vein to preserve maximal length of the vessels. Hindlimb bone was separated using a Gigli Saw. Hindlimbs were cannulated through the femoral artery and underwent immediate arterial flush with up to 200-400 mL heparinized-saline solution (10,000 units of heparin per 1L of saline), until venous outflow was clear. Additionally, the femoral vein was cannulated for measurement and collection of venous outflow samples.

Limbs were divided into two groups, control (N=3) and experimental SNC80 (N=5). All hindlimbs were connected to the ENCORE[®] head for perfusion via the cannulated femoral artery. The femoral vein was cannulated for obtaining venous samples. Hindlimbs were perfused with oxygenated Krebs-Henseleit (KH) at ambient temperature. Arterial and venous pO₂ were assessed hourly using a commercial blood gas analyzer (ALB1000 by Radiometer, Brea, California USA). Oxygen consumption was calculated from A-V differences obtained from arterial and venous blood gas readings. After 3 hours of perfusion (baseline), torpor inducing compound SNC80 (44.96 µg/ml) was administered via the femoral artery to the experimental group (SNC80) (N=5) and an equal volume of the sham vehicle to the control limbs (N=3) followed by and additional 3 hours of perfusion (drug). The hind limbs were then removed from the perfusor and flushed with fresh KH perfusate and replaced into the perfusor containing fresh KH solution (flush). The perfusion then resumed for an additional 6 hours. The hindlimbs were weighed pre and post perfusion. Biopsy tissue samples were taken at baseline (pre-ENCORE[®]), and before and after the administration of the SNC80 compound for RNA-Seq analysis and histological evaluation.

Biopsies for histopathology were obtained superficially from the rectus femoris muscle bundle immediately after amputation and then 24 hours after the perfusion period. Biopsies were stored in 10% formalin for 24 hours and then stored for 3 days in 70% ethanol and embedded in paraffin. The samples were sectioned into 5µm slides and stained with hematoxylin and eosin (H&E) to evaluate tissue morphology, intracellular distance, muscle bundle area, and Periodic Acid-Schiff to evaluate glycogen content. Histological analysis was performed by visualizing slides under a light microscope (AM Scope T670Q, Irvine, California) and capturing images using a high-definition camera (AM Scope Model# MU2003B1, Irvine, California). From each slide, 10 digital images from randomly selected fields were obtained at 10x magnification. The images were de-identified and evaluated for intracellular distance, muscle bundle area and glycogen content by two independent reviewers using Image J software (v.1.51.).

Statistical analysis

All graphing and statistical analyses were performed with Prism 9 (GraphPad Software Inc., La Jolla, CA, USA) with a (two-sided) significance level of 0.05. Statistical tests and corrections for multiple comparisons are described in each figure panel.

Data availability statement

The proteomics data reported in this paper have been deposited in the Mass Spec Interactive Virtual Environment, MassIVE, (accession no. MSV000091410). All other data are available within the paper and its Supplementary Information files.

Software availability statement

A custom Python script was used to automatically count heart beat in *Xenopus* tadpoles. This code can be accessed on GitHub: https://github.com/k-shcherb/heart_seg .

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Author Contributions

R.N., M.Su., and D.E.I. conceived of and directed the work. M.M.S., H.F., E.G., A.K., S.K., M.P.O., K.She., and T.L. performed *Xenopus* studies with guidance from R.N. and M.Lev. R.M. developed and maintained engineered components for *Xenopus* assays. H.F., M.M.S., A.K., M.P.O., and T.T. analyzed *Xenopus* data. T.L. and M.Lew. performed MALDI-TOF experiments and analyzed the data. B.C., K.P., and Z.I. performed in vitro and Organ Chip studies and analyzed data with guidance from A.M.S. S.M. and K.Shc. designed, assembled, and maintained custom microscopy equipment for Organ Chip imaging. K.Shc. analyzed imaging data from tadpole and in vitro

experiments. S.L. oversaw compound management and material transfers. J.M. designed novel chemical entities. Porcine heart experiments were conceived and directed by R.V. and L.B. and experiments were designed and interpreted by R.V., L.B., and K.A. Porcine heart data analysis was performed by R.V., L.B., K.A., E.C., R.L., I.C., Z.M., and D.D.C. Porcine heart data was acquired by K.A., E.C., R.L., I.C., and Z.M. Porcine limb studies were conceived and directed by R.V., L.B., E.W., and T.P. Porcine limb surgeries were performed by J.Y., O.P., and J.S. Porcine limb sampling and data analysis were performed by R.V., L.B., J.P., D.W., Z.H., C.H. and J.C.H. M.M.S., B.C., K.P., Z.I., T.L., K.A., R.N. and D.E.I. drafted the manuscript and all authors reviewed and provided critical input for the manuscript.

Competing Interests

D.E.I. is a founder, board member and Scientific Advisory Board chair of, and holds equity in, Emulate Inc. M.Le., R.N., M.M.S., E.G., T.T., and D.E.I. are inventors on a relevant patent application held by Harvard University (PCT/US2021/012626). R.N., E.G., M.Le., and D.E.I. hold equity in Unravel Biosciences, Inc.; R.N. and D.E.I. are members of its board of directors; M.Le. and D.E.I. are members of its scientific advisory board; and R.N. and E.G. are current employees of the company. R.V., L.B., K.A., E.C., R.L., I.C., and Z.M. are employees of Vascular Perfusion Solutions. D.E.I. is a member of its scientific advisory board. The remaining authors declare no competing interests.

Supplemental Figure Legends

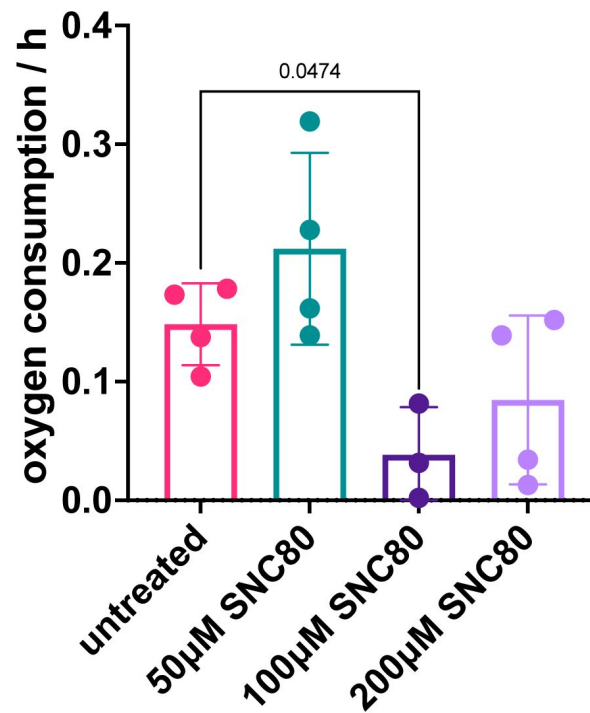


Figure S1.

Metabolic measurements in SNC80-treated *Xenopus* embryos.

Oxygen consumption rate in *Xenopus* embryos treated with SNC80 (n=3-4 replicates/group). Statistical comparisons were performed using a Brown-Forsythe and Welch ANOVA test with Dunnett correction for multiple comparisons between untreated and each treatment group. Data represent the mean±SD.

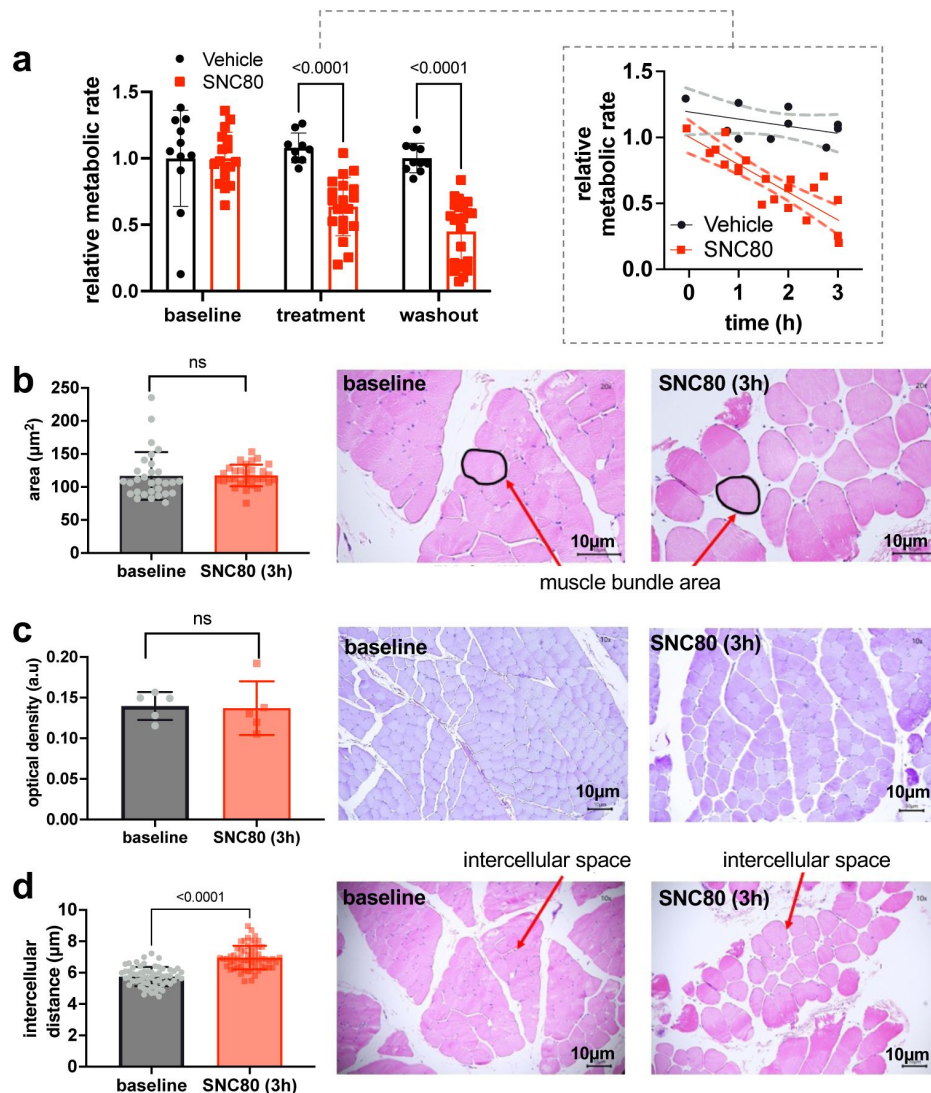


Figure S2.

Preservation of whole porcine limbs with SNC80.

a. The relative metabolic rate at baseline is not statistically significantly different between control ($n=3$) and SNC80 ($n=5$) injected limbs. Metabolic rate is reduced during the treatment and washout phases for SNC80 relative to controls. **b.** Mean muscle bundle area between T0 before SNC80 administration and at T3 after SNC80 administration and histograms of limb biopsy (20x magnification) at T0 before SNC80 administration and T3 after SNC80 administration. Muscle bundle area is not affected by the administration of SNC80. **c.** Optical density between T0 before SNC80 administration and at T3 after SNC80 administration and histograms of limb biopsy (20x magnification) at T0 before SNC80 administration and T3 after SNC80 administration. Optical density is not affected by the administration of SNC80. **d.** Mean intracellular distance during T0 before SNC80 administration versus T3 after SNC80 administration and histograms of limb biopsy (10x magnification) at T0 before SNC80 administration and T3 after SNC80 administration. Red arrows indicate intracellular distance measured. Intracellular distance increases after the administration of SNC80. Statistical comparisons were performed using a mixed-effects analysis with a Sidak's test for multiple comparisons (a) and Welch's t-test (b-d). Data represent the mean $\pm$ SD.

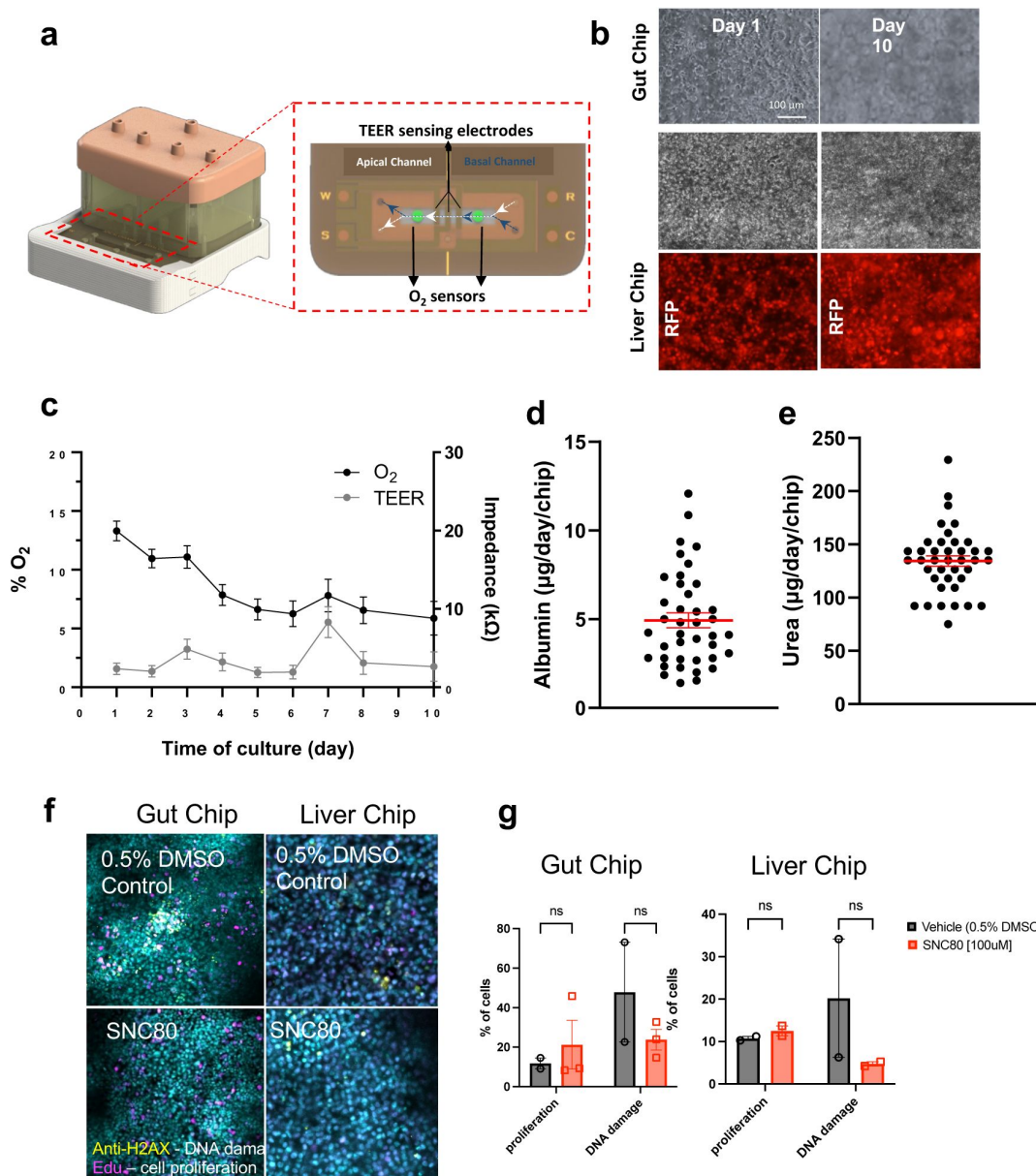


Figure S3.

Characterization of human Gut and Liver Chips with integrated sensing technology.

a. Illustration of the sensor-integrated OOC unit connected to reservoir pod for automatic operation of fluid flow using pneumatic pressure (left). Inset shows a schematic of the top and bottom views of the sensor-chip-PCB compartment showing locations of TEER electrodes and O₂ sensors in the apical and basal channels. Four O₂ sensing spots were added to the channels - two in the top and two in the bottom channels for across the chip measurement. Electrode connectors are necessary for remote TEER sensing. **b.** Phase contrast images of Caco-2 cells seeded on sensor-chip at day 1, and 10 of culture (top) and Huh7 cells seeded on sensor-chip at day 1 and 10 (middle). Fluorescent images of liver chip at day 1 and 10 of the transduced nuclear marker (bottom). **c.** Percentage of O₂ level (black) and TEER impedance values (gray) on sensor Liver Chip during 10 days of culture. **d.** Average albumin production of Liver Chips at day of experiment start (day 7-10 of culture). **e.** Average urea production of Liver Chips at day of experiment start (day 7-10 of culture). **f.** Representative immunofluorescence images of the sensor Gut Chip (left) and Liver Chip (right) epithelium stained with DNA damage marker (H2AX) and nuclei (DAPI) for vehicle control and SNC80 treated tissues after 7 days recovery. **g.** Quantitative analysis of proliferation and DNA damage stains on Gut and Liver Chips after 48h of treatment and 7-day recovery to 100μM SNC80 (2-3 biological replicates/group/marker). Statistical comparisons were performed using multiple unpaired t-tests for each marker with FDR correction. Data represent the mean±SD.

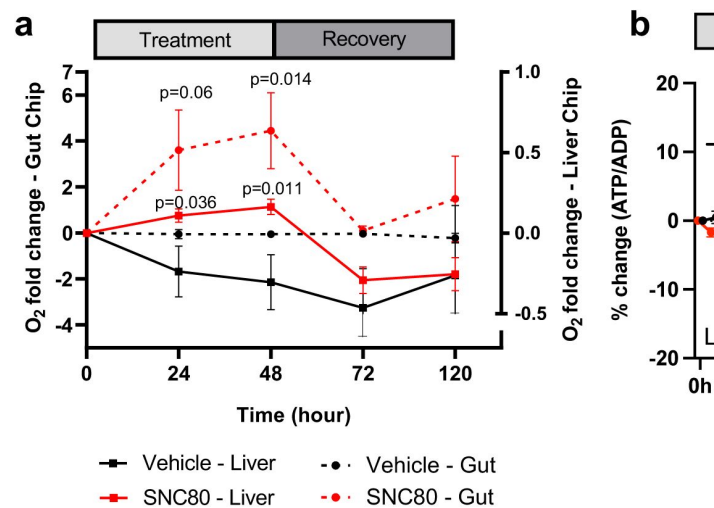


Figure S4.

SNC80 treatment in human Organ Chips.

a. Monitoring changes in O₂ levels in human Gut (dotted lines; n=6 Chips/group) or Liver (straight lines; n=15-16 Chips/group) Chips throughout the treatment with 100 μ M SNC80 (0-48h) and recovery (48-120h) versus vehicle control Chip.

b. Graph showing the quantitative drop in ATP/ADP ratio of Liver Chip treated for 48h with 100 μ M SNC80 or Vehicle control as well as 2 days recovery after drug washout (n=5 control and n=3 SNC80 Chips). Statistical comparisons were performed using multiple unpaired t-tests for each timestamp with multiple comparison between control and treated cells. Data represent the mean $\pm$ SD.

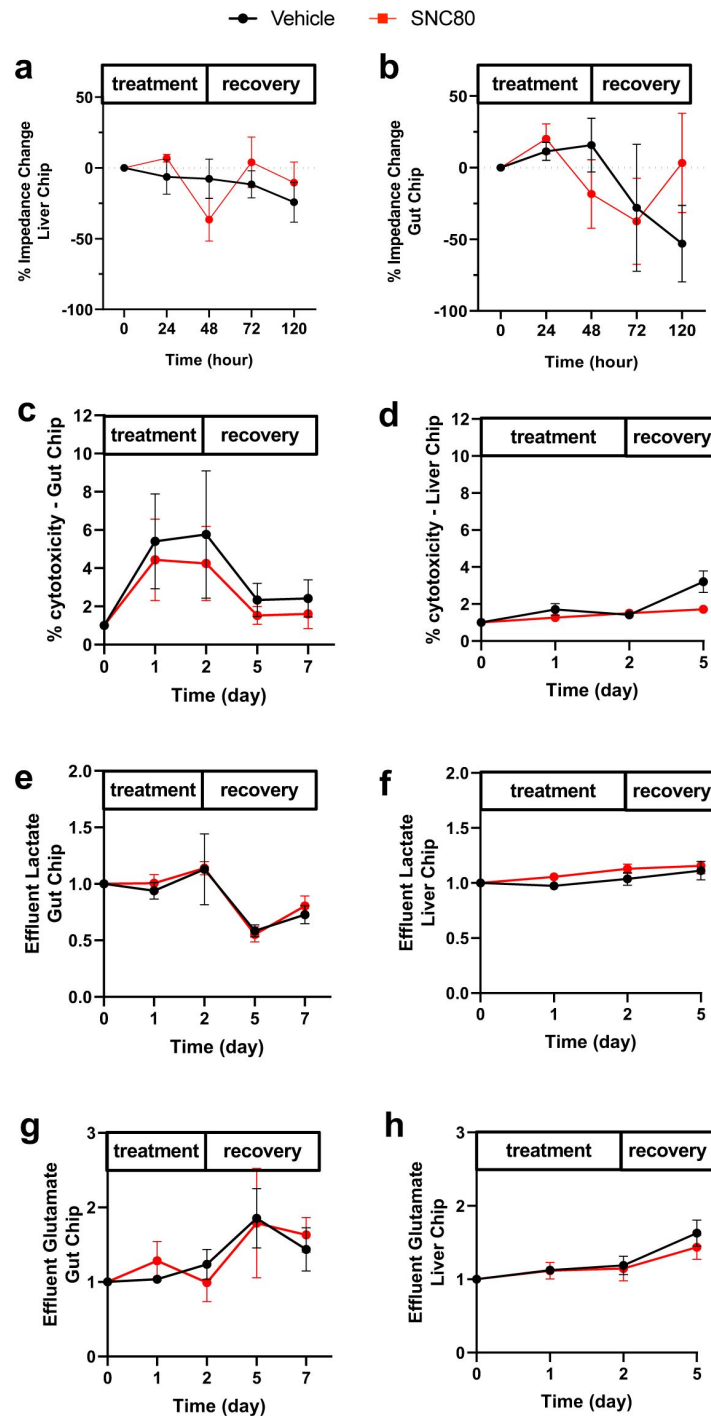


Figure S5.

Metabolic measurements of Gut and Liver Chip systems treated with SNC80.

a. Monitoring the percentage change in barrier function of Gut (n=6 Chips/group) or **b.** Liver Chip (n=9 Chips/group) with time during treatment with 100 μ M SNC80 (0-48h) and recovery (48-120h) versus vehicle control Chip. **c.** Cytotoxicity measured for the Gut and **d.** Liver Chips during treatment and recovery. **e.** Graph depicting the amount of lactate measured in effluents of Gut and **f.** Liver Chips post recovery (day 7 for Gut and day 5 for Liver) for SNC80-treated (red) or vehicle control (black) Chips. **g.** Graph showing the glutamate measured in effluents of Gut or **h.** Liver Chips during SNC80 (red) or vehicle (black) treatment and post recovery (day 7). For each measurement, statistical comparisons were performed using multiple unpaired t-tests at each timepoint and FDR correction for multiple comparisons. Data represent the mean $\pm$ SD.

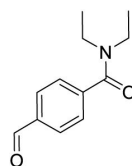
		Before recirculating					
		Pulse (bpm)	Rate	Rhythm	P waves	QRS waves	T waves
SNC80	P014	n/a	normocardic	irregular	no	no	n/a
	P015	n/a	tachycardic	irregular	no	no	n/a
	P019	n/a	n/a	n/a	n/a	n/a	n/a
	P023	12 bpm	bradycardic	irregular	no	no	n/a
Vehicle	P016	n/a	tachycardic	irregular	no	no	n/a
	P020	n/a	tachycardic	irregular	no	no	n/a
	P024	n/a	tachycardia	irregular	no	no	n/a
		After defibrillation					
		Pulse (bpm)	Rate	Rhythm	P waves	QRS waves	T waves
SNC80	P014	132 bpm	tachycardic	regular	yes	yes	T wave elevation
	P015	96bpm	normocardic	irregular	yes	yes	Normal T waves
	P019	54 bpm	bradycardic	irregular	some	yes	Inconsistent T wave depression
	P023	96 bpm	normocardic	regular	no	no	Unable to assess (supraventricular tachycardia)
Vehicle	P016	144 bpm	tachycardic	regular	yes	yes	Slightly abnormal ST segment depression
	P020	132 bpm	tachycardic	regular	no	yes	T wave depression
	P024	138 bpm	tachycardic	regular	some	yes	T wave abnormal (P and T wave overlap)

Table S1.

Clinical interpretation of ECG in SNC80 and vehicle treated hearts.

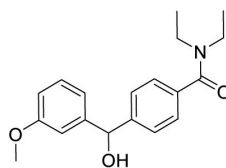
Supplemental Synthesis Description

Step 1: Synthesis of intermediate *N,N*-diethyl-4-formylbenzamide



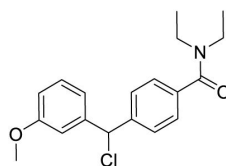
4-formylbenzoic acid (10 g, 66.61 mmol) and HATU (30.39 g, 79.93 mmol) were dissolved in DMF (500 mL) at 15 °C under N₂. The mixture was then warmed to 30 °C and stirred for 1 h. At this juncture, the mixture was cooled to 15 °C and DIEA (34.43 g, 266.43 mmol, 46.41 mL) and diethylamine (7.31 g, 99.91 mmol, 10.29 mL) were added dropwise together. The reaction was allowed to warm to rt and stirred for 12 h. The reaction mixture was diluted with H₂O (2000 mL), and then extracted with ethyl acetate (5000 mL x 3). The combined organic layers were washed with brine (1000 mL), dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO₂, Petroleum ether/Ethyl acetate = 3/1 to 1/1) to give *N,N*-diethyl-4-formylbenzamide (57 g, 277.71 mmol, 83.38% yield) as a yellow solid. LC-MS *m/z* = 206.1 [M+H]⁺.

Step 2: Synthesis of intermediate *N,N*-diethyl-4-[hydroxy(3-methoxyphenyl)methyl]benzamide



To a solution of *N,N*-diethyl-4-formylbenzamide (8 g, 38.98 mmol) in THF (320 mL) was dropwise added bromo-(3-methoxyphenyl)magnesium (1 M, 46.77 mL) at 0 °C under N₂. The reaction was stirred at 20 °C for 2 h, at which time it was diluted with H₂O (1000 mL), and then extracted with dichloromethane (500 mL x 3). The combined organic layers were washed with brine (300 mL), dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give a residue. The residue was purified by column chromatography (SiO₂, Petroleum ether/Ethyl acetate = 3/1 to 1/1) to give *N,N*-diethyl-4-[hydroxy-(3-methoxyphenyl)methyl]benzamide (6 g, 19.15 mmol, 6.14% yield) as a white solid. LC-MS *m/z* = 314.1 [M+H]⁺.

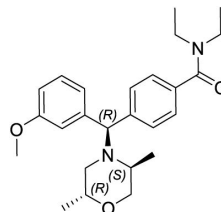
Step 3: Synthesis of intermediate 4-(chloro(3-methoxyphenyl)methyl)-*N,N*-diethylbenzamide



To a solution of *N,N*-diethyl-4-[hydroxy-(3-methoxyphenyl)methyl]benzamide (11.75 g, 37.49 mmol) in CHCl₃ (110 mL) was added dropwise HCl (12 M, 421.80 mL). The resulting mixture was stirred at 20 °C for 12 h, at which time it was diluted with H₂O (50 mL), and then extracted with

dichloromethane (30 mL x 3). The combined organic layers were washed with brine (30 mL), dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give 4-[chloro-(3-methoxyphenyl)methyl]-*N,N*-diethyl-benzamide (23 g, 69.31 mmol, 92.43% yield) as a yellow oil.

Step 4: Synthesis of WB3



To a mixture of 4-[chloro-(3-methoxyphenyl)methyl]-*N,N*-diethyl-benzamide (23 g, 69.31 mmol) and (2*R*,5*S*)-2,5-dimethylmorpholine (15.97 g, 138.62 mmol) in ACN (230 mL) was added K₂CO₃ (19.16 g, 138.62 mmol). The resulting mixture was stirred at 80 °C for 4 days. The reaction mixture was filtered, and the mother liquor was directly concentrated. The residue was purified by column chromatography (SiO₂, Petroleum ether/Ethyl acetate = 5/1 to 2/1). The residue was further purified by preparative-HPLC (column: UniSil 10-120 C18 70×250mm; mobile phase: [water(FA)-ACN]; gradient:20%-50% B over 20 min) and the appropriate fractions were lyophilized. The residue was again further purified by preparative-HPLC (column: Phenomenex luna C18 (250*70mm,10 um); mobile phase: [water (FA)-ACN]; gradient:10%-50% B over 35 min) and the appropriate fractions were lyophilized to give WB3 (3.3 g, 8.03 mmol, 11.59% yield) as a white solid. LC-MS *m/z* = 296.1 [M+H]⁺.

¹H NMR (400 MHz, CDCl₃) δ = 7.47 (d, *J* = 7.8 Hz, 2H), 7.34 - 7.27 (m, 3H), 6.88 (dd, *J* = 2.0, 8.3 Hz, 1H), 6.75 (d, *J* = 7.6 Hz, 1H), 6.72 - 6.67 (m, 1H), 5.32 (s, 1H), 3.81 (s, 3H), 3.76 - 3.66 (m, 2H), 3.64 - 3.45 (m, 2H), 3.44 - 3.24 (m, 3H), 2.61 (dd, *J* = 2.0, 11.6 Hz, 1H), 2.57 - 2.48 (m, 1H), 1.82 (dd, *J* = 9.5, 11.6 Hz, 1H), 1.24 (br d, *J* = 0.8 Hz, 3H), 1.12 (d, *J* = 6.1 Hz, 6H), 1.06 (d, *J* = 6.3 Hz, 3H).

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Reviewer #1 (Public Review):**Summary:**

In this study of metabolism using *Xenopus*, explanted porcine hearts and limbs, and human organs-on-chips, Sperry et al studied the ability of WB3 to slow metabolism and mobility. The group developed WB3, an analog of SNC80, void of SNC80's delta-opioid receptor binding capacity, and studied its metabolic impact. The authors concluded that SNC80 and its analog WB3 can induce "biostasis" and produce a hypometabolic state which holds promise for prolonging organ viability in transplant surgery as well as other potential clinical benefits.

Strengths:

This study also opens new avenues for therapeutic possibilities in areas such as trauma, acute infection, and brain injuries. The overall methodology is acceptable, but certain concerns should be addressed.

Weaknesses:

1. In cardiac and renal transplantation, cold preservation in ice remains a common practice for transporting explanted hearts to donors which remains a cheap and easily accessible way of preserving organs. While ex-vivo mechanical circulatory platforms have been developed and are increasingly being utilized to prolong organ viability, cold preservation remains widely used. The authors perfused explanted hearts with oxygenated perfusion preservation devices at subnormothermic temperatures (20-23C) which is even much lower than routinely used in clinical cardiopulmonary bypass scenarios (28-32C) (in the discussion, the authors allude to SNC80's possible "protective effect" in cardiac bypass). It is unclear how much of the hypometabolic state is related to WB3 administration versus hypothermia. The study will benefit from a comparison of WB3 administration and hypothermia in *Xenopus*, explanted porcine organs versus cold preservation alone to show distinction in biostasis parameters.

2. The authors selected SNC80 based on a literature survey where it was identified based on its ability to induce hypothermia and protect against the effects of spinal cord ischemia in rodents. While this makes sense, were other drugs (eg. Puerarin) considered? The induction

of hypothermia and spinal cord protective effect of SNC80 may be multifactorial and not necessarily related to its biostatic effects as the authors describe. Please provide some more context into the background of SNC80.

3. In most of the models, the primary metric that the authors utilize to characterize metabolic activity is oxygen consumption, which is a somewhat limited indicator. For instance, this does not provide any information, however, on anaerobic metabolic activity. In addition, the ATP/ADP ratio was found to decrease in the organ chips where SNC80 was utilized, but similar findings were not presented for the other models.

4. The authors should provide a more detailed explanation of SNC80's mechanisms of interaction with proteins related to transmembrane transport, mitochondrial activity, and metabolic processes. What is the impact of SNC80 on mitochondrial function, particularly ATP production and mitochondrial respiration? Are there changes in mitochondrial membrane potential, electron transport chain activity, or oxidative phosphorylation? In this context, the authors discuss the potential role of NCX1 as a binding target for SNC80 and its various mechanisms in slowing metabolism. However, no experiments have been done to confirm this binding in the present study. Co-immunoprecipitation studies using appropriate antibodies against SNC80 and NCX1 should be considered to demonstrate their direct binding. Additionally, surface plasmon resonance (SPR) or isothermal titration calorimetry (ITC) experiments could be employed to quantify the binding affinity between SNC80 and NCX1, providing further evidence of their interaction. These experiments would elucidate the binding mechanism between SNC80 and NCX1 and reveal more information on the mechanism of action for SNC80.

5. The manuscript notes that histological analysis was conducted, but it seems that only example images are provided, such as Figure 4f. Quantified histological data would provide a more thorough understanding of tissue integrity.

6. Some of the points mentioned in the discussion and conclusion are rather strong and based on possible associations such as SNC80's potential vasodilatory capacity conferring a cardioprotective effect, and ability to reversibly suppress metabolism across different temperatures and species. Please tone this down and stay limited to the organs studied. Further, the reversibility of the findings may be more objectively assessed by biomarkers with decreased immunofluorescence in response to ischemia such as troponin I for the heart and albumin for the liver. Additionally, an investigation of proteins involved in inflammation, hypoxia, and key cell death pathways using immunohistochemistry analysis can better describe the impact of treatment on apoptosis/necroptosis.

7. What could be the underlying cause of the observed increase in intercellular spacing after SNC80 administration in porcine limbs which also seems to be evident in the heart histology samples? This seems to be more prominent in the SNC80 compared to the vehicle group.

8. In the Discussion section, it would be valuable to provide a concise interpretation of the lipidomic data, particularly explaining how changes in acylcarnitine and cholesterol ester levels may relate to tadpole metabolism, hibernation, or other biological processes.

9. What are the limitations or disadvantages of the study? Does SNC80 possess any immunomodulatory properties that might affect the outcomes of organ transplantation? Are there specific organs for which SNC80 may not be a suitable preservation agent, and if so, what are the reasons behind this?

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Reviewer #2 (Public Review):**Summary:**

This manuscript titled "Identification of pharmacological inducers of a reversible hypometabolic state for whole organ preservation" reports the effects of delta opioid receptor activator SNC80 and its modified analog WB3 with ~1,000 times less delta opioid receptor binding activity on metabolic state.

Strengths:

This is an interesting study with potentially broad implications for organ preservation.

Weaknesses:

There are several limitations that raise concerns.

1. The authors developed an analog of a known delta opioid receptor activator SNC80 with three orders of magnitude lesser binding with the delta opioid receptor WB3. This will likely reduce the undesirable effects of SNC80 while preserving the metabolic slowing needed for organ preservation. Yet, most experiments were done with SNC80, not the superior modification, WB3, shown in only a limited set of experiments, Figure 3.
2. The heart is one of the most challenging organs to preserve, and some experiments are done to establish the metabolic effects of SNC80. However, the biodistribution study, shown in Figure 2, conspicuously omitted the heart.
3. I do not understand the design of the electrophysiology and contractility experiments with the porcine hearts. How did you defibrillate the hearts after removal and establishing perfusion? Lines 173-175 on Page 7 state: "After defibrillation with epinephrine, the P and QRS waveforms were visible in ECGs from 3 of 4 SNC80-treated hearts (Table S1), suggesting that those hearts regain atrial and ventricular polarization." Please clarify. Defibrillation is done with an electric shock. Also, please show the ECG recordings to support your conclusions about "polarization." What did you mean by "polarization"? Depolarization? Repolarization? Or resting potential. To establish a normal physiological state, please show ECG waveforms and present data on basic ECG characteristics: heart rate, PQ and QT intervals, and P and QRS durations. I recommend perfusion of the porcine heart with WB3, not only SNC80.
4. Pathology data also raises concerns. The histology images shown in Figure 4f are not quantified, and they show apparently higher levels of tissue disruption in SNC80-treated tissue vs vehicle-treated. The test (lines 169-171) confirms this concern: "In some hearts treated with SNC80, greater waviness of muscle fibers was observed, possibly indicating a state of muscle contraction." It will be helpful to measure markers of apoptosis and necrosis and to apply TTC viability staining.
5. The apparent state of contracture suggests a higher degree of myocardial damage and a high intracellular calcium level in SNC80-treated hearts. The authors suggested that the sodium-calcium exchanger NCX is a possible target of SNC80 and could be responsible for the "hypometabolic state." However, NCX1 is critically important in the extrusion of cytosolic Ca^{2+} during the diastolic phase. Failure to remove excessive calcium and restore ionic homeostasis would lead to calcium overload and heart failure.
6. I am surprised the authors did not consider using the gold standard assay for measuring mitochondrial function in cells by the Seahorse Cell Mito Stress Test.

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Reviewer #3 (Public Review):

In this manuscript, Sperry and colleagues identify SNC80 as a compound that can slow metabolism and mimic hibernation, thereby prolonging tissue viability in organ transplantation and cardiovascular disease settings. Overall, the use of varied and relevant model systems is a strength of this study.

The authors perform a literature search to identify SNC80 as a promising hit. However, the details of the literature search, a list of other potential hits, and the criteria for identification of SNC80 are not described. The hypometabolic effect of SNC80 exposure is well-characterized in the *Xenopus* model. Furthermore, the authors show that SNC80 localises to the brain, but do not discuss several studies that have pointed to convulsions induced by exposure to high doses of SNC80, and whether this would be apparent in the *Xenopus* studies. The authors have promising data on the WB3 morpholino that retains or even improves on the hypometabolism phenotype of SNC80 while likely not retaining delta opioid activity. However, this is not functionally demonstrated. Moreover, WB3 is not used in any of the other assays and models used in the study. In the setting of cardiac transplant surgery, co-administration of SNC80 reduces metabolic activity and inflammation, although it is unclear if there is an improvement in recovery of organ function due to SNC80. The reversible induction of hypometabolic status is also demonstrated in two different organ chips. These models could identify the differential response of epithelial cells and vascular cells to drug perfusion, but the authors have mostly focused on the former. Finally, the authors identify specific targets for the hypometabolic effect of SNC80, which is a valuable resource for other screening studies and can form the basis for future work.

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